BLOCKCHAIN BASED E-VOTING SYSTEM

A project submitted in partial fulfillment of the requirements for the degree of

B.Sc.INFORMATION TECHNOLOGY

SUBMITTED BY

N. FAREEDHUL FARZANA(22ITA11)

&

R.SRITHA(22ITA35)

UNDER THE GUIDANCE OF

Mrs.R.LAKSHMI, MCA.,MPHIL.,B.Ed.



DEPARTMENT OF INFORMATION TECHNOLOGY

E.M.G.YADAVA WOMEN'S COLLEGE MADURAI-625014

(An Autonomous Institution–Affiliated to Madurai Kamaraj University)

(Re-accredited(3rd Cycle) with Grade A^+ and CGPA 3.51 by NAAC)

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DEPARTMENT OF INFORMATION TECHNOLOGY

BONAFIDE CERTIFICATE

This is to certify that the project work done at "**BLOCKCHAIN BASED E-VOTING SYSTEM Using Python**" is a work done by **N.FAREEDHUL FARZANA (22ITA11) & R.SRITHA (22ITA35)** submitted to the department of B.Sc.,(Information Technology), E.M.G. Yadava Women's College , Madurai, In partial fulfillment for the award of Bachelor of Information Technology during the period of 2022-2025.

Submitted for the Viva-Voice exam held at E.M.G Yadava Women's College, Madurai-14 on……..

HEAD OF THE DEPARTMENT

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

We hereby declare that the project work entitled "**BLOCKCHAIN BASED E-VOTING SYSTEM**" submitted to the department of B.Sc.(Information Technology), E.M.G. Yadava Women's College for the award of Bachelor of Information Technology is a record of an original work done by us under the guidance of Mrs.R.LAKSHMI, MCA.,MPHIL.,B.Ed, Assistant Professor, E.M.G. YADAVA WOMEN'S COLLEGE, Madurai and this project work have not performed the basic for the award of any Degree or diploma/associateship/fellowship and similar project if any.

Signature of the candidate 1

Signature of the candidate 2

Place:

Date:

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BLOCKCHAIN BASED E-VOTING SYSTEM

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ABSTRACT

The **Blockchain Based E-Voting System** is an innovative and secure digital voting platform designed to enhance election integrity using blockchain technology. Developed with Python and Flask, this system ensures transparency, security, and efficiency by leveraging Aadhaar-based user authentication, OTP verification, and video recording for additional voter verification. The platform enables users to register seamlessly by verifying their Aadhaar ID and phone number, ensuring only legitimate voters participate. During the voting process, an integrated video recording system captures voter activity, preventing fraud and impersonation. Votes are securely stored on a custom blockchain, making them immutable and resistant to tampering, while real-time vote counting provides instant election updates. An admin panel allows election authorities to verify registered users and oversee the voting process, ensuring compliance with election protocols. By integrating these features, the **Blockchain Based E-Voting System** offers a reliable, tamper-proof, and user-friendly alternative to traditional voting methods, reinforcing trust in digital elections and promoting democratic transparency.

OBJECTIVES

The main objective of our project is:

- **Secure and Transparent Voting Process:** Leverage blockchain technology to provide a tamper-proof, decentralized voting system that ensures election integrity and eliminates fraud.
- **Voter Authentication and Anonymity:** Implement cryptographic verification methods to authenticate eligible voters while maintaining voter anonymity.
- **Decentralized and Immutable Ledger:** Store votes on a decentralized blockchain ledger, ensuring transparency, immutability, and accessibility for auditing.
- **Efficiency and Accessibility:** Enable a seamless and user-friendly voting experience, making elections accessible to remote and physically challenged voters.
- **Real-time Vote Tracking and Results:** Provide real-time vote tracking while ensuring results are verifiable, quick, and secure without external manipulation.

INTRODUCTION

In today's fast-paced, technology-driven world, the need for secure, transparent, and efficient voting systems is more crucial than ever. Blockchain technology has revolutionized the way we ensure trust and integrity in digital transactions, making it an ideal solution for modernizing election processes. The **Blockchain Based E-Voting System** is an innovative solution designed to empower voters by providing a secure, decentralized, and transparent voting platform. By leveraging blockchain technology, this system ensures that election data remains tamper-proof, enabling seamless, fraud-resistant voting through a decentralized network.

The core of this system lies in its ability to provide a secure and transparent voting process by leveraging blockchain technology. It ensures that every vote is recorded immutably, preventing tampering and fraudulent activities while maintaining voter anonymity. With the rise of digital transformation, online voting systems have become essential for enhancing accessibility and efficiency in elections. Blockchain-based e-voting systems enable a decentralized and trustless voting environment, eliminating the need for intermediaries and reducing the risk of manipulation. By integrating cryptographic security measures, this system offers a streamlined and verifiable election process, ensuring fairness and integrity in democratic voting.

A blockchain-based e-voting system leverages decentralized ledger technology to securely record, verify, and manage votes while ensuring transparency and immutability. It uses cryptographic algorithms and consensus mechanisms to authenticate voters, prevent double voting, and safeguard election integrity. Blockchain technology ensures that each vote is encrypted and stored in a tamper-proof manner, eliminating risks associated with traditional electronic voting systems. While some blockchain-based voting systems are designed as standalone platforms, others can be integrated into existing governmental or organizational infrastructures to enhance security and accessibility. By decentralizing the voting process, this system minimizes reliance on intermediaries, reducing vulnerabilities and fostering trust among voters.

Nowadays, blockchain-based e-voting systems are being integrated into various digital platforms, including government election systems, corporate decision-making processes, and decentralized organizations. As the demand for secure and transparent voting mechanisms grows, blockchain technology offers a revolutionary approach to ensuring trust and efficiency in electoral processes. The ability to cast votes remotely using a secure and verifiable system enhances voter convenience, allowing individuals to participate in elections from the comfort of their homes without concerns about fraud or manipulation.

Blockchain-Based E-Voting System is an advanced and secure voting platform developed using blockchain technology to revolutionize the electoral process. As digital transformation continues to expand, the need for a transparent, tamper-proof, and decentralized voting system has become essential. This system leverages blockchain to ensure integrity, anonymity, and trust in elections, enabling a seamless and fraud-resistant voting experience. By utilizing decentralized ledgers and cryptographic security, the Blockchain-Based E-Voting System enhances the reliability of elections, making the voting process more efficient, verifiable, and accessible for all users.

The Blockchain-Based E-Voting System incorporates several key features that emphasize its security, transparency, and efficiency. One of the standout aspects is its ability to record and verify votes on a tamper-proof blockchain ledger, ensuring that each vote remains immutable and traceable. This decentralized approach prevents fraud and unauthorized modifications, enhancing election credibility. Additionally, the system employs cryptographic techniques to maintain voter anonymity while securing data integrity, ensuring a confidential and trustworthy voting process. Together, these features promote a more transparent, reliable, and technologically advanced electoral system.

In addition to its core voting functionalities, the Blockchain-Based E-Voting System integrates advanced features that enhance usability and efficiency. The system provides an automated vote tallying mechanism, eliminating manual counting errors and accelerating result computation. Furthermore, it incorporates a secure voter authentication process using cryptographic verification, ensuring that only eligible voters can participate. To improve engagement and transparency, the system offers a real-time vote tracking feature, allowing users to confirm that their votes have been securely recorded on the blockchain. By blending security, efficiency, and user-friendliness, the Blockchain-Based E-Voting System sets a new standard for modern electoral processes.

MODULE DESCRIPTION

- User Registration Module
- Login and Authentication Module
- Voting Module
- Blockchain Module
- Results Module
- Admin Panel Module

User Registration Module:

- **Secure User Authentication:** The voter registration module ensures that only eligible users can register by implementing a secure authentication process. Users must provide valid identification details, which are verified before granting access to the voting system.
- **Unique Voter ID Assignment:** Each registered voter is assigned a unique voter ID, which prevents duplicate or fraudulent registrations. This ID is securely stored in both the database and the blockchain, ensuring transparency and immutability.
- **Data Encryption & Privacy Protection:** All voter information, including personal details and authentication credentials, is encrypted using robust security protocols such as AES-256 and RSA encryption. This ensures that sensitive voter data remains confidential and protected from unauthorized access.
- **Blockchain Integration for Transparency:** The voter registration details are stored on the blockchain, ensuring that once registered, the voter information remains unaltered and secure. This decentralized approach eliminates the risk of data manipulation or tampering.
- **User-Friendly Interface:** The registration process is designed to be simple and accessible for all users. A guided interface helps voters complete the registration efficiently, reducing errors and enhancing usability.

Login and Authentication Module:

- **Secure Login System:** The authentication and authorization module ensures that only verified users can access the e-voting system. It employs a multi-factor authentication (MFA) process, requiring users to verify their identity through passwords, OTP (One-Time Password), or biometric authentication.

- **Role-Based Access Control (RBAC):** The system assigns different access levels based on user roles (voter, admin, election officer). Voters can only cast their votes, while admins can manage elections, ensuring a secure and controlled environment.
- **Blockchain-Based Identity Verification:** The module leverages blockchain technology to store and validate user identities, ensuring transparency and preventing unauthorized modifications. Once registered, the voter's credentials are stored securely and cannot be altered.
- **Encryption & Data Protection:** User authentication data is encrypted using advanced cryptographic techniques such as SHA-256 hashing for passwords and end-to-end encryption for sensitive information. This guarantees data security and prevents unauthorized access.
- **Session Management & Auto-Logout:** To enhance security, the system implements session timeouts and automatic logout features, preventing unauthorized access in case of user inactivity.
- **Tamper-Proof Access Logs:** All authentication attempts, successful or failed, are logged on the blockchain, making it easy to track suspicious activities and ensure accountability.

Voting Module:

- **Video Recording for Verification:** The system records a short video of the voting process to verify the voter's identity and ensure transparency. This helps in preventing fraudulent activities and maintaining election integrity.
- **Real-Time Vote Encryption:** Votes are securely recorded on the blockchain with end-to-end encryption, preventing unauthorized access and tampering.
- **Authentication & Authorization:** Before casting a vote, the system verifies the voter's identity using biometric authentication, OTP verification, or a unique voter ID. Only authorized users can proceed with the voting process.
- **Tamper-Proof Records:** Once a vote is cast, it is permanently stored on the blockchain, making it immutable and resistant to fraud.
- **Automated Confirmation & Audit Trail:** After a successful vote, the system provides a confirmation message and generates a unique transaction ID for audit purposes, ensuring complete transparency.

Blockchain Module:

- **Secure Data Storage:** All voting data, including user credentials, vote records, and timestamps, are stored on the blockchain, ensuring data integrity and preventing unauthorized alterations.
- **Decentralized Ledger:** The blockchain distributes stored data across multiple nodes, eliminating a single point of failure and enhancing security.
- **Immutability & Transparency:** Once data is recorded, it cannot be altered or deleted, maintaining a transparent and tamper-proof record of all transactions.
- **Encryption & Privacy:** Uses cryptographic techniques like SHA-256 hashing and asymmetric encryption to protect voter identities and sensitive data. Only authorized entities can access decrypted information.
- **Smart Contracts for Automation:** Implements self-executing smart contracts to validate, store, and verify votes without manual intervention, ensuring a seamless and fraud-resistant voting process.
- **Backup & Redundancy:** The distributed nature of blockchain ensures that even if one node fails, data remains intact and accessible across other nodes, preventing data loss.

Results Module:

- **Real-Time Vote Counting:** The system automatically tallies votes as they are cast, providing instant and accurate results without manual intervention.
- **Data Visualization:** Results are displayed using bar charts, pie charts, and tables, highlighting the number of votes each candidate has received, along with overall voter turnout.
- **Candidate Ranking:** Votes are sorted in ascending or descending order to easily identify leading and trailing candidates.
- **Blockchain Verification:** Each vote is validated through the blockchain, ensuring transparency and preventing any manipulation of results.
- **Voter Participation Insights:** The system generates analytics on voter turnout, time-based voting trends, and demographics, helping election organizers understand participation patterns.
- **Exportable Reports:** Election results and analytics can be exported in various formats (CSV, PDF) for official records and further analysis.

Admin panel Module :

- **User Management:** The admin can monitor registered users, verify voter identities, and manage access permissions.
- **Election Management:** Enables administrators to create, update, and configure election settings, including candidate details, voting timeframes, and eligibility criteria.
- **Vote Monitoring:** Provides real-time tracking of votes cast, ensuring transparency and preventing fraudulent activities.
- **System Logs & Security Alerts:** Displays blockchain transaction logs and notifies the admin of any suspicious activity or unauthorized access attempts.
- **Data Analytics & Reports:** Generates reports on voter participation, election trends, and final results, which can be exported for record-keeping.
- **User-Friendly Interface:** Designed for seamless navigation, allowing administrators to efficiently manage and oversee the voting system.

SYSTEM STUDY

EXISTING SYSTEM:

Traditional electronic voting systems rely on centralized databases and conventional authentication methods, making them vulnerable to security threats such as data manipulation, hacking, and unauthorized access. Many existing systems store votes in centralized servers, increasing the risk of tampering and reducing voter trust in the electoral process. Additionally, these systems often depend on outdated encryption techniques that may not provide sufficient protection against modern cyber threats.

Another significant limitation of current e-voting platforms is the lack of transparency and verifiability. Voters cannot independently verify whether their votes have been counted correctly, leading to concerns over election integrity. Moreover, traditional e-voting systems may not ensure anonymity while maintaining auditability, which is a critical requirement for a fair election process.

To address these shortcomings, **Blockchain-Based E-Voting Systems** provide a decentralized, secure, and transparent solution. By leveraging blockchain technology, votes are encrypted, recorded immutably, and verified through consensus mechanisms, ensuring trust, security, and tamper-proof elections.

DISADVANTAGE

- **Centralized Vulnerability:** Traditional e-voting systems rely on centralized databases, making them susceptible to hacking, data breaches, and manipulation.
- **Lack of Transparency:** Voters cannot verify whether their votes have been recorded and counted accurately, leading to distrust in the system.
- **Security Risks:** Existing systems often use outdated encryption methods, making them prone to cyberattacks, unauthorized access, and vote tampering.
- **Limited Accessibility:** Many e-voting systems require physical presence at polling stations or rely on complex authentication methods, restricting accessibility for remote or disabled voters.
- **High Maintenance Costs:** Centralized servers and proprietary software require constant monitoring and maintenance, increasing operational costs.

PROPOSED SYSTEM:

The proposed **Blockchain-Based E-Voting System** is designed to address the limitations of traditional voting systems by leveraging blockchain technology to ensure transparency, security, and efficiency. Unlike conventional e-voting solutions that rely on centralized servers, this system decentralizes vote storage, making it tamper-proof and verifiable. Each vote is recorded on the blockchain as an immutable transaction, preventing unauthorized modifications and ensuring election integrity.

One of the key advantages of this system is its **end-to-end encryption and voter authentication** mechanisms, which safeguard voter identities while maintaining anonymity. The system employs smart contracts to automate vote counting, reducing human intervention and minimizing the risk of manipulation. Additionally, a **real-time results and analytics dashboard** enhances transparency, allowing voters and election authorities to monitor the process securely.

By integrating blockchain, the proposed system ensures **fair, transparent, and fraud-resistant elections**, empowering voters with verifiable proof of their participation while reducing administrative overhead. This approach not only modernizes the voting process but also builds public trust in digital democracy.

ADVANTAGE

- **Enhanced Security & Transparency:** Blockchain ensures that all votes are securely recorded in an immutable ledger, preventing tampering and fraud.
- **Decentralization:** Unlike traditional voting systems, which rely on a central authority, blockchain-based voting eliminates single points of failure, making elections more trustworthy.
- **Voter Anonymity & Privacy:** Cryptographic encryption protects voter identities while ensuring that each vote is counted accurately and securely.
- **Tamper-Proof Vote Storage:** Once a vote is cast, it cannot be altered or deleted, ensuring election integrity.
- **Real-Time Vote Counting:** Smart contracts automate vote tallying, reducing delays and minimizing human errors.
- **Remote & Accessible Voting:** Blockchain-based e-voting enables voters to securely participate in elections from anywhere, improving accessibility for citizens with mobility issues or those living abroad.

- **Cost-Effective & Efficient:** Eliminates the need for physical ballots, reducing election costs and administrative workload.
- **Auditability & Verifiability:** Voters can verify that their votes were recorded correctly while maintaining anonymity, increasing public trust in the election process.

FEASIBILITY STUDY

Technical Feasibility

1. Blockchain Implementation:

- Evaluate blockchain frameworks such as **Hyperledger Fabric**, **Ethereum (private network)**, or **custom-built blockchain** for secure and decentralized vote recording.
- Ensure efficient **consensus mechanisms** (Proof of Authority or Proof of Stake) to maintain a balance between security and performance.
- Test blockchain scalability and transaction throughput to handle a large number of votes simultaneously.

2. Cryptographic Security:

- Implement **asymmetric encryption (RSA, ECC)** to protect voter identities.
- Utilize **hashing algorithms (SHA-256)** for secure vote storage and verification.
- Ensure end-to-end encryption for data transmission between voters and the blockchain.

3. Smart Contracts:

- Develop **self-executing smart contracts** to automate voter verification, vote counting, and result declaration.
- Test contract security to prevent vulnerabilities like **reentrancy attacks** and **double voting issues**.

4. User Authentication & Voter Registration:

- Integrate **multi-factor authentication (MFA)** to ensure only eligible voters can cast a vote.
- Verify voter identities using **biometric authentication (fingerprint, facial recognition)** or **government-issued IDs**.

5. Integration & Compatibility:

- Ensure cross-platform support for web and mobile-based voting applications.
- Develop **APIs** to facilitate seamless interaction between the Block chain and the front-end system.
- Test performance under different network conditions to ensure smooth accessibility

6. **System Performance & Optimization:**

- Evaluate blockchain latency and optimize smart contract execution time.
- Implement **load balancing** techniques to handle high traffic during elections.
- Ensure the system can recover from **node failures** without data loss.

Economic Feasibility

1. Cost Analysis:

Development Costs:

- ✓ Blockchain infrastructure setup (e.g., private Ethereum network, Hyperledger Fabric).
- ✓ Smart contract development and security audits.
- ✓ Web and mobile application development for voter interaction.
- ✓ Database integration and secure cloud storage for voter authentication.

Hardware & Server Costs:

- ✓ Server infrastructure to host the voting platform.
- ✓ Blockchain node setup and maintenance.
- ✓ Security measures (encryption tools, firewall, DDoS protection).

Maintenance Costs:

- ✓ Continuous security updates and bug fixes.
- ✓ Server maintenance and bandwidth costs.
- ✓ Smart contract modifications for future elections.

2. Return on Investment (ROI):

Cost Savings:

- ✓ Eliminates the cost of physical polling stations, ballot papers, and manual vote counting.
- ✓ Reduces human errors and fraud, saving money on security measures.
- ✓ Faster result processing reduces administrative costs.

Monetization Strategies:

- ✓ Offering the e-voting system to **government agencies, universities, and private organizations**.
- ✓ Providing **premium security features** (e.g., biometric verification, audit trails).
- ✓ Licensing the platform as a **Software-as-a-Service (SaaS)** for organizations conducting internal elections.

Organizational Feasibility

1. **Stakeholder Support:**

- Gather input from key stakeholders, such as **election commissions, government bodies, cybersecurity experts, and voters**, to ensure alignment with **electoral regulations and security standards**.
- Conduct **surveys or focus groups** with **voters, polling officials, and IT professionals** to assess interest, concerns, and specific requirements for implementing a blockchain-based e-voting system.

2. **Training and Adoption:**

- Develop **training materials and user guides** to educate **election officials, administrators, and voters** on how to use the blockchain-based e-voting system securely and efficiently.
- Conduct **mock elections or trial runs** before official implementation to ensure **smooth adoption and identify potential challenges**.
- Implement a **phased rollout strategy**, starting with **small-scale local elections** before expanding to **larger governmental elections**, ensuring proper **testing, security validation, and user confidence**.

Legal and Ethical Feasibility

1. **Privacy and Security**:

- Ensure **compliance with data protection laws** (e.g., **GDPR, CCPA, and election-specific regulations**) by implementing **strong cryptographic encryption** for voter data and ballot information.

- Utilize **blockchain's decentralized nature** to ensure **tamper-proof** and **immutable** voting records.

- Implement **multi-factor authentication (MFA)** and **secure digital identities** to prevent unauthorized voting.

- Conduct **regular security audits** and **penetration testing** to identify and mitigate vulnerabilities, preventing data breaches and cyberattacks.

2. **Ethical Considerations**:

- **Eliminating Voting Bias & Fairness:**

  - Ensure **equal voting rights** by designing the system to **prevent discrimination** based on factors like **location, device access, or digital literacy**.
  - Implement **fair vote counting algorithms** and ensure that the **blockchain ledger remains immutable**, preventing any unfair alterations or bias in vote recording.

- **Transparency & Trust in Decision-Making:**

  - Maintain **public auditability** by allowing independent election authorities or third parties to **verify blockchain records** without compromising voter anonymity.
  - Provide **clear and accessible voting guidelines** so that all users understand how the system operates, fostering trust.
  - Publish **regular reports on system performance, vote counting accuracy, and any security updates** to ensure transparency in elections.

- **Ensuring Ethical Voter Data Handling:**

  - nforce **strict voter privacy policies**, ensuring that **personal identities remain confidential** while keeping voting records verifiable.
  - Avoid using voter data for **unauthorized purposes** and ensure that the system adheres to **strict ethical and legal data protection standards**.

Cultural Feasibility

1.User Acceptance:

- **Conduct pilot tests** with diverse voter groups to identify usability issues and gather feedback.
- **Design an accessible interface** that accommodates various literacy levels and ensures ease of use for all voters, including those with disabilities.
- **Educate voters on blockchain technology** through awareness campaigns, tutorials, and government outreach programs to build trust and confidence in the system.

2. Localization:

- **Support multiple languages and regional dialects** to ensure voters from different backgrounds can easily understand and interact with the system.
- **Customize the user experience** by adapting election procedures, system instructions, and ballot designs to reflect local cultural and electoral norms.
- **Ensure legal and procedural compliance** with regional voting laws while maintaining the transparency and integrity of blockchain-based voting.

SYSTEM SPECIFICATION

HARDWARE SPECIFICATION

- RAM:8GB
- Storage:512 SSD
- Processor:12th Gen Intel(R) Core (TM) i3-1215U 1.20 GHz

SOFTWARE REQUIREMENTS

- O/S : Windows 10.
- Language : Python
- Front End : Anaconda Navigator – Spyder

SOFTWARE ENVIRONMENT

OPERATING SYSTEM

Windows 11 serves as the backbone of the Voice Assistance for System Control project, providing a stable and feature-rich environment for executing voice commands and automating system tasks. As a modern operating system, Windows 11 is equipped with advanced speech recognition capabilities, improved AI-driven features, and enhanced system performance, making it an ideal choice for voice-controlled automation. The project is developed and executed using Spyder from Anaconda, which is fully compatible with Windows 11, ensuring the smooth execution of Python scripts required for voice processing and system control. One of the key advantages of using Windows 11 is its seamless support for speech recognition and text-to-speech (TTS) functionalities. The operating system integrates well with Python libraries such as speech-recognition, which converts spoken language into text, and pyttsx3, which enables the assistant to respond with synthesized speech. These features allow users to interact with their computer through voice commands efficiently. Furthermore, Windows 11 provides built-in APIs such as Microsoft Speech API (SAPI), which enhances the accuracy and responsiveness of voice-based interactions.

The automation and system control capabilities of Windows 11 allow the voice assistant to perform various tasks such as opening applications, adjusting system settings, managing files and folders,

and automating repetitive operations. This is achieved through Python libraries like pyautogui, which simulates keyboard and mouse actions, and the os module, which enables direct execution of system-level commands. Additionally, Windows PowerShell and Command Prompt can be leveraged within Python scripts to execute administrative tasks, ensuring deeper control over the operating system. Security and user access control are also significant aspects of Windows 11 that contribute to the project. The operating system ensures that voice commands requiring administrative permissions are executed safely while preventing unauthorized system modifications. Features such as User Account Control (UAC) and Windows Defender Security help maintain a secure execution environment, preventing malicious code from affecting the system.

Furthermore, Windows 11's optimized resource management and AI-driven enhancements contribute to the efficient functioning of the voice assistant. The OS effectively manages CPU and memory resources, ensuring smooth performance even when handling multiple background processes. Its support for multi-threading and efficient task scheduling helps in executing voice commands with minimal latency, enhancing the user experience.

Overall, Windows 11 provides the necessary infrastructure for building a reliable and efficient voice assistant capable of controlling system functions through voice commands. Its support for modern AI technologies, seamless Python integration, robust security measures, and automation capabilities make it a powerful platform for developing intelligent voice-based system control applications.

LANGUAGE DESCRIPTION

PYTHON

Python is one of those rare languages that can claim to be both simple and powerful. You will find yourself pleasantly surprised to see how easy it is to concentrate on the solution to the problem rather than the syntax and structure of the language you are programming in. The official introduction to Python is an easy-to-learn, powerful programming language. It has efficient high-level data structures and a simple but effective approach to object-oriented programming. Python's elegant syntax and dynamic typing, together with its interpreted nature, make it an ideal language for scripting and rapid application development in many areas on most platforms. I will discuss most of these features in more detail in the next section.

Features of Python

- **Simple**

Python is a simple and minimalistic language. Reading a good Python program feels almost like reading English, although it is very strict English! This pseudo-code nature of Python is one of its greatest strengths. It allows you to concentrate on the solution to the problem rather than the language itself.

- **Easy to Learn**

As you will see, Python is extremely easy to get started with. Python has an extraordinarily simple syntax, as already mentioned.



- **Free and Open Source**

Python is an example of a *FLOSS* (Free/Libré and Open Source Software). In simple terms, you can freely distribute copies of this software, read its source code, make changes to it, and use pieces of it in new free programs. FLOSS is based on the concept of a community that shares knowledge. This is one of the reasons why Python is so good - it has been created and is constantly improved by a community who just want to see a better Python.

- **High-level Language**

When you write programs in Python, you never need to bother about low-level details such as managing the memory used by your program, etc.

- **Portable**

Due to its open-source nature, Python has been ported to (i.e. changed to make it work on) many platforms. All your Python programs can work on any of these platforms without requiring any changes at all if you are careful enough to avoid any system-dependent features.

- **Interpreted**

A program written in a compiled language like C or C++ is converted from the source language, i.e., C or C++, into a language that is spoken by your computer (binary code, i.e., 0s and 1s) using a compiler with various flags and options. When you run the program, the linker/loader software copies the program from the hard disk to memory and starts running it. Python, on the other hand, does not need compilation to binary. You just run the program directly from the source code. Internally, Python converts the source code into an intermediate form called bytecodes, translates this into the native language of your computer, and then runs it.

All this makes using Python much easier since you don't have to worry about compiling the program, making sure that the proper libraries are linked and loaded, etc. This also makes your Python programs much more portable since you can just copy your Python program onto another computer, and it just works!

- **Object Oriented**

Python supports procedure-oriented programming as well as object-oriented programming. In procedure-oriented languages, the program is built around procedures or functions which are nothing but reusable pieces of programs. In object-oriented languages, the program is built around objects that combine data and functionality. Python has a very powerful but simplistic way of doing OOP, especially when compared to big languages like C++ or Java.

- **Extensible**

If you need a critical piece of code to run very fast or want to have some piece of algorithm not to be open, you can code that part of your program in C or C++ and then use it from your Python program.

- **Embeddable**

You can embed Python within your C/C++ programs to give *scripting* capabilities for your program's users.

- **Extensive Libraries**

The Python Standard Library is huge indeed. It can help you do various things involving regular expressions, documentation generation, unit testing, threading, databases, web browsers, CGI, FTP,

email, XML, XML-RPC, HTML, WAV files, cryptography, GUI (graphical user interfaces), and other system-dependent stuff. Remember, all this is always available wherever Python is installed. This is called the Batteries Included philosophy of Python. Besides the standard library, there are various other high-quality libraries that you can find in the Python Package Index.

IMPORTING LIBRARIES

FLASK:

Flask is a lightweight and flexible Python web framework used to build web applications. It provides essential tools and features for handling user requests, rendering templates, and managing data flow between the frontend and backend. In a blockchain-based e-voting system, Flask plays a crucial role in enabling secure and efficient user interactions, such as voter registration, login authentication, and vote submission. Its simple syntax and modular structure make it an ideal choice for developing scalable and maintainable web applications.

SQLITE:

SQLite is a lightweight, embedded database system that stores data in a single file, making it an efficient choice for applications requiring local storage. In a blockchain-based e-voting system, SQLite is used to store voter information, vote records, and election results. Its integration with Python allows easy data retrieval, manipulation, and storage operations without requiring a complex database setup. Since SQLite is serverless, it ensures a hassle-free implementation, making it a practical choice for small-scale and decentralized voting systems.

HASHLIB:

Hashlib is a Python library that provides various cryptographic hashing functions, such as SHA-256, ensuring data integrity and security. In an e-voting system, hashlib is used to securely hash votes before adding them to the blockchain, preventing tampering or unauthorized modifications. By using cryptographic techniques, hashlib helps maintain transparency and trust in the voting process, ensuring that each vote remains unaltered once cast.

JSON:

The JSON library in Python enables seamless data formatting and storage, making it a crucial component of a blockchain-based system. Since blockchain transactions involve storing and retrieving structured data, JSON helps encode and decode blockchain records efficiently. It

facilitates communication between different components of the voting system, ensuring data consistency and interoperability.

TIME:

Time is a built-in Python module used for managing timestamps and scheduling events. In a blockchain-based e-voting system, it plays a vital role in recording the exact time of each vote transaction, ensuring chronological order and preventing fraudulent activities. By timestamping each block in the chain, the system maintains a verifiable and transparent voting history.

MATPLOTLIB:

Matplotlib is a popular data visualization library in Python that enables the creation of charts and graphs. In an e-voting system, matplotlib is used to generate bar charts representing voting results, helping administrators and users visualize the election outcome. By providing clear and insightful graphical representations, it enhances decision-making and improves voter transparency.

RANDOM:

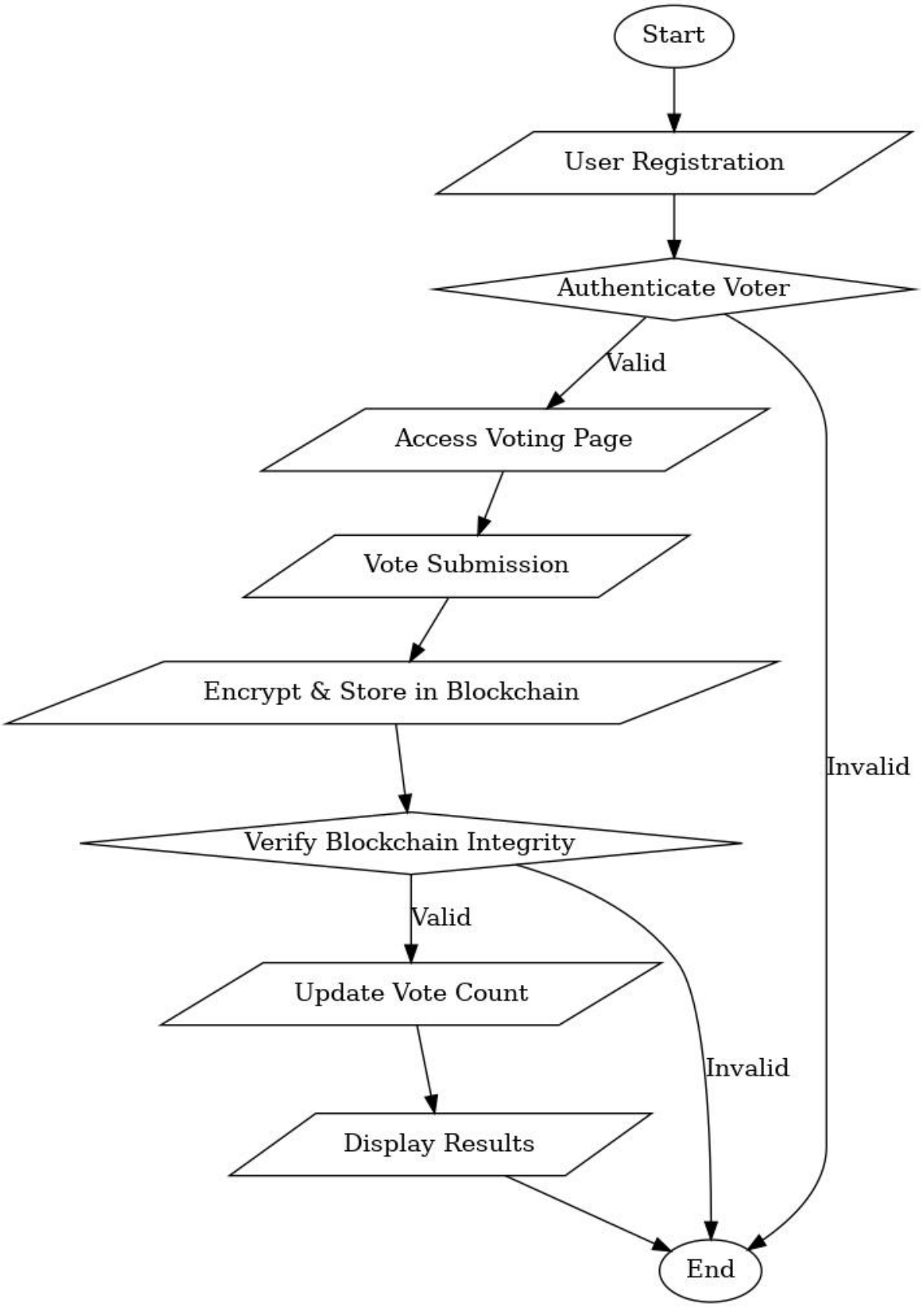
The Random module is used for generating pseudo-random numbers, which can be useful in security-related applications such as cryptographic salt generation. In an e-voting system, it can be employed to generate unique voter IDs or randomize ballot order, enhancing privacy and security.

OS:

The OS module provides functionalities for interacting with the operating system, such as file handling and directory management. In a blockchain-based e-voting system, it can be used to manage database file paths and dynamically access system resources, ensuring efficient data storage and retrieval.

By integrating these libraries, the blockchain-based e-voting system ensures a seamless, secure, and efficient voting process. Each library plays a specific role in handling web interactions, securing vote transactions, storing data, visualizing results, and maintaining overall system integrity.

FLOW DIAGRAM



Flow of the System :

User Registration

1. Users register with their credentials.
2. Data is stored in the **SQLite database**.

Login & Authentication

1. Users log in with their registered credentials.
2. The system verifies the user against the database.

Voting Process

1. Verified users access the voting page.
2. They select a candidate and submit their vote.
3. The vote is **hashed and stored** in the **blockchain** and **SQLite** database.

Blockchain Validation

1. Each vote is **hashed and linked** to the previous vote.
2. The blockchain ensures **tamper-proof** voting records.

Result Calculation

1. Votes are counted and stored in the **SQLite database**.
2. Matplotlib is used to display **bar charts** for visual representation.

Admin Panel

1. Admin verifies votes and checks for anomalies.
2. The final result is displayed.

SYSTEM TESTING

TESTING:

System testing is a crucial stage in the implementation of a **Blockchain-Based E-Voting System**, ensuring that the system works accurately and efficiently before live deployment. Testing involves executing the program with the intent of detecting errors. A well-designed test case has a high probability of identifying errors, ensuring the system's reliability and robustness.

System testing assumes that if all parts of the system function correctly, the overall goal will be successfully achieved. A series of tests are conducted before the system is ready for **user acceptance testing**. Any engineered product can be tested in two primary ways:

- Functional testing to verify that each function operates correctly.
- Internal testing to ensure all components interact as expected.

UNIT TESTING:

Unit testing focuses on individual modules within the **Blockchain-Based E-Voting System** to ensure their correctness. This process verifies the smallest unit of software design, known as **module testing**. Each module is tested separately during the programming phase.

Validation checks include:

- Ensuring correct data entry format.
- Validating user inputs.
- Debugging and fixing errors efficiently.

INTEGRATION TESTING:

Integration testing ensures seamless interaction between modules. Errors such as data loss across interfaces, adverse module interactions, or incorrect functional combinations are identified. This systematic approach ensures the overall system performance meets expectations.

Types of integration testing:

- **Top-down integration testing**
- **Bottom-up integration testing**

TESTING TECHNIQUES/STRATEGIES:

White box testing:

White box testing examines the internal logic and structure of the **Blockchain-Based E-Voting System** to ensure:

- All independent paths within a module are tested.
- Each decision point executes all possible outcomes.
- Loops are tested at their boundaries and within operational ranges.

Black Box Testing:

Black box testing focuses on the system's external behavior without considering its internal logic. This ensures:

- Identification of incorrect or missing functions.
- Detection of interface errors.
- Validation of external database access.
- Performance optimization.
- Identification of initialization and termination errors.

VALIDATION TESTING:

After black box testing, software validation begins to confirm that the **Blockchain-Based E-Voting System** meets expected performance and user requirements. Validation succeeds when the system functions as reasonably expected by the customer.

USER ACCEPTANCE TESTING:

User acceptance testing ensures that the system meets user needs and expectations. The **Blockchain-Based E-Voting System** undergoes rigorous testing with real users, ensuring adaptability and functionality before full deployment.

OUTPUT TESTING:

Output testing ensures the correctness of system-generated outputs in both **screen display** and **hard copy format**. User requirements guide the output validation process, confirming that the system provides accurate and well-formatted results.

ANNEXURE

SAMPLE CODE

APP.PY (USING FLASK WEB FRAMEWORK)

```
from flask import Flask, render_template, request, redirect, url_for, flash, session,
jsonify,send_from_directory
from flask_sqlalchemy import SQLAlchemy
from flask_mail import Mail, Message
import cv2
import time
import random
import os
import hashlib
import json
from werkzeug.utils import secure_filename
app = Flask(__name__)

# Database Configuration
app.config['SQLALCHEMY_DATABASE_URI'] = 'sqlite:///evoting.db'
app.config['SQLALCHEMY_TRACK_MODIFICATIONS'] = False
app.config['SECRET_KEY'] = 'your_secret_key'

db = SQLAlchemy(app)

# Email Configuration for OTP
app.config['MAIL_SERVER'] = 'smtp.gmail.com'
app.config['MAIL_PORT'] = 587
app.config['MAIL_USE_TLS'] = True
app.config['MAIL_USERNAME'] = 'your-email@gmail.com'
app.config['MAIL_PASSWORD'] = 'your-email-password'

mail = Mail(app)

UPLOAD_FOLDER = "uploads"
if not os.path.exists(UPLOAD_FOLDER):
    os.makedirs(UPLOAD_FOLDER)
```

```
VIDEO_UPLOAD_FOLDER = 'video_records'  # Video recordings
os.makedirs(VIDEO_UPLOAD_FOLDER, exist_ok=True)
app.config['VIDEO_UPLOAD_FOLDER'] = VIDEO_UPLOAD_FOLDER

# Blockchain Storage
BLOCKCHAIN_FILE = "blockchain.json"

# Ensure blockchain file exists
if not os.path.exists(BLOCKCHAIN_FILE):
    with open(BLOCKCHAIN_FILE, "w") as f:
        json.dump([], f)

# Blockchain Structure
class Blockchain:
    def __init__(self):
        self.chain = self.load_blockchain()

    def load_blockchain(self):
        """Load existing blockchain from file"""
        with open(BLOCKCHAIN_FILE, "r") as f:
            return json.load(f)

    def save_blockchain(self):
        """Save blockchain to file"""
        with open(BLOCKCHAIN_FILE, "w") as f:
            json.dump(self.chain, f, indent=4)

    def create_block(self, voter_id, candidate):
        """Create a new block for the vote"""
        previous_hash = self.chain[-1]['hash'] if self.chain else "0"
        vote_data = f"{voter_id}-{candidate}-{time.time()}"
        block_hash = hashlib.sha256(vote_data.encode()).hexdigest()

        block = {
            "index": len(self.chain) + 1,
```

```
            "voter_id": voter_id,
            "candidate": candidate,
            "timestamp": time.time(),
            "previous_hash": previous_hash,
            "hash": block_hash,
        }

        self.chain.append(block)
        self.save_blockchain()
        return block

    def verify_integrity(self):
        """Verify blockchain integrity"""
        for i in range(1, len(self.chain)):
            if self.chain[i]['previous_hash'] != self.chain[i - 1]['hash']:
                return False
        return True

blockchain = Blockchain()



# User Model
class User(db.Model):
    id = db.Column(db.Integer, primary_key=True, autoincrement=True)
    first_name = db.Column(db.String(50), nullable=False)
    last_name = db.Column(db.String(50), nullable=False)
    voter_id = db.Column(db.String(10), unique=True, nullable=False)
    aadhaar_id = db.Column(db.String(12), unique=True, nullable=False)
    phone_number = db.Column(db.String(15), unique=True, nullable=False)
    otp = db.Column(db.String(6), nullable=True)
    aadhaar_card_path = db.Column(db.String(255), nullable=True)
    voted = db.Column(db.Boolean, default=False)  # ✅ Added to track voting status
    video_path = db.Column(db.String(255), nullable=True)  # ✅ Added to store video path

# Voting Record Model
class VotingRecord(db.Model):
```

```
    id = db.Column(db.Integer, primary_key=True, autoincrement=True)
    voter_id = db.Column(db.String(10), db.ForeignKey('user.voter_id'), nullable=False)  # Foreign
Key
    candidate = db.Column(db.String(100), nullable=False)
    video_path = db.Column(db.String(255), nullable=False)  # Store video path
    timestamp = db.Column(db.DateTime, default=db.func.current_timestamp())  # Store vote time

    user = db.relationship('User', backref=db.backref('voting_records', lazy=True))

@app.route("/vote", methods=["GET"])
def vote():
    if "voter_id" not in session:
        flash("Please log in first.", "danger")
        return redirect(url_for("login"))

    user = User.query.filter_by(voter_id=session["voter_id"]).first()

    if not user:
        flash("User not found.", "danger")
        return redirect(url_for("login"))

    if user.voted:
        flash("You have already voted.", "danger")
        return redirect(url_for("dashboard"))

    # Start Timer (5 minutes)
    session["expiry_time"] = time.time() + 300

    return render_template("voting_page.html", voter_id=session["voter_id"])
    if "voter_id" not in session:
     return redirect(url_for("login"))
     return render_template("voting_page.html")
@app.route("/submit_vote", methods=["POST"])
def submit_vote():
    print("Received a request to submit vote")  # Debugging
```

```
if "video" not in request.files or "candidate" not in request.form:
    print("Missing video or candidate!")  # Debugging
    return jsonify({"error": "Missing video file or candidate selection."}), 400

video_file = request.files["video"]
candidate = request.form["candidate"]

if video_file.filename == "":
    print("Empty video file received!")  # Debugging
    return jsonify({"error": "Empty video file uploaded."}), 400

if "voter_id" not in session:
    return jsonify({"error": "User not logged in."}), 400

voter_id = session["voter_id"]  # Get voter ID from session

# Generate a unique filename (voter_id.mp4)
video_filename = f"{voter_id}.mp4"
video_path = os.path.join(app.config['VIDEO_UPLOAD_FOLDER'], video_filename)

print(f"Saving video to: {video_path}")  # Debugging
video_file.save(video_path)

# Store vote in blockchain
blockchain.create_block(voter_id, candidate)
print(json.dumps(blockchain.chain, indent=4))



# Store vote in the database
user = User.query.filter_by(voter_id=voter_id).first()
if user:
    user.video_path = video_path  # Save video path to User table
    user.voted = True  # Mark user as voted
    new_vote = VotingRecord(voter_id=voter_id, candidate=candidate, video_path=video_path)
    db.session.add(new_vote)
    db.session.commit()
```

```
    return jsonify({"message": "Vote submitted successfully!"})
@app.route("/verify_blockchain", methods=["GET"])
def verify_blockchain():
    if blockchain.verify_integrity():
        return jsonify({"message": "Blockchain integrity verified. No tampering detected!"})
    else:
        return jsonify({"error": "Blockchain has been tampered with!"}), 400
@app.route("/login", methods=["GET", "POST"])
def login():
    if request.method == "POST":
        voter_id = request.form.get("voterId")
        phone_number = request.form.get("phoneNumber")

        user = User.query.filter_by(voter_id=voter_id, phone_number=phone_number).first()

        if not user:
            flash("User not found. Please register first.", "danger")
            return redirect(url_for("register"))

        session["voter_id"] = voter_id
        flash("Login successful! Redirecting to voting page.", "success")
        return redirect(url_for("vote"))  # ✅ Fixed this line
    print("Session voter_id:", session.get("voter_id"))  # Debugging


    return render_template("login.html")

@app.route("/logout")
def logout():
    session.pop("voter_id", None)
    flash("Logged out successfully.", "success")
    return redirect(url_for("login"))

@app.route("/", methods=["GET", "POST"])
def register():
```

```
if request.method == "POST":
    voter_id = request.form.get("voterId", "").strip()
    phone_number = request.form.get("phoneNumber", "").strip()
    first_name = request.form.get("first_name", "").strip()
    last_name = request.form.get("last_name", "").strip()
    aadhaar_id = request.form.get("aadhaar_id", "").strip()
    aadhaar_card_path = request.form.get("aadhaar_card_path", "").strip()


    if not first_name:
        flash("First name is required!", "danger")
        return redirect(url_for("register"))

    if not last_name:
        flash("Last name is required!", "danger")
        return redirect(url_for("register"))

    if "aadhaarCard" not in request.files:
        flash("Please upload your Aadhaar card.", "danger")
        return redirect(url_for("register"))

    aadhaar_card = request.files["aadhaarCard"]
    if aadhaar_card.filename == "":
        flash("No Aadhaar card selected!", "danger")
        return redirect(url_for("register"))

    if User.query.filter_by(voter_id=voter_id).first():
        flash("Voter ID already registered. Try logging in.", "danger")
        return redirect(url_for("login"))

    if User.query.filter_by(phone_number=phone_number).first():
        flash("Phone number already registered. Try logging in.", "danger")
        return redirect(url_for("login"))

    # Save Aadhaar Card File
    aadhaar_path = os.path.join(UPLOAD_FOLDER, aadhaar_card.filename)
```

```
        aadhaar_card.save(aadhaar_path)

        # Generate OTP
        otp = str(random.randint(100000, 999999))
        session[phone_number] = otp

        # Save new user in database
        new_user = User(
            voter_id=voter_id,
            phone_number=phone_number,
            first_name=first_name,
            last_name=last_name,
            aadhaar_id=aadhaar_id,
            aadhaar_card_path=aadhaar_path,
            otp=otp
        )
        db.session.add(new_user)
        db.session.commit()

        flash("Registration successful! OTP sent to your email.", "success")
        return redirect(url_for("login"))

    return render_template("register.html")

@app.route("/send_otp", methods=["POST"])
def send_otp():
    phone_number = request.form.get("phoneNumber")

    if not phone_number:
        return jsonify({"error": "Phone number is required!"}), 400

    # Generate 6-digit OTP
    otp = str(random.randint(100000, 999999))
    session[phone_number] = otp  # Store OTP in session

    # Print OTP to terminal only (no email)
```

```
    print(f"✅ OTP for {phone_number}: {otp}")

    return jsonify({"message": "OTP generated and printed in terminal."})
@app.route("/verify_otp", methods=["POST"])
def verify_otp():
    data = request.json
    phone_number = data.get('phoneNumber')
    otp_entered = data.get('otp')

    # Retrieve the correct OTP from session
    correct_otp = session.get(phone_number)

    if not correct_otp:
        return jsonify({"error": "No OTP found for this phone number. Please request a new OTP."}),
400

    if otp_entered == correct_otp:
        return jsonify({"message": "OTP verified successfully!"}), 200
    else:
        return jsonify({"error": "Invalid OTP. Please try again."}), 400
@app.route("/send_login_otp", methods=["POST"])
def send_login_otp():
    voter_id = request.form.get("voterId")
    phone_number = request.form.get("phoneNumber")

    if not voter_id or not phone_number:
        return jsonify({"error": "Voter ID and Phone Number are required!"}), 400

    # Check if voter ID and phone exist in the database
    user = User.query.filter_by(voter_id=voter_id, phone_number=phone_number).first()
    if not user:
        return jsonify({"error": "User not found. Please register first."}), 400

    # Generate OTP
    otp = str(random.randint(100000, 999999))
    session[phone_number] = otp  # Store OTP in session
```

```
    # Print OTP in terminal (for testing)
    print(f"✅ Login OTP for {phone_number}: {otp}")

    return jsonify({"message": "OTP sent. Check your terminal."})


@app.route("/verify_login_otp", methods=["POST"])
def verify_login_otp():
    data = request.json
    phone_number = data.get('phoneNumber')
    otp_entered = data.get('otp')

    # Retrieve the correct OTP from session
    correct_otp = session.get(phone_number)

    if not correct_otp:
        return jsonify({"error": "No OTP found for this phone number. Please request a new OTP."}),
400

    if otp_entered == correct_otp:
        return jsonify({"message": "OTP verified successfully! You can now log in."}), 200
    else:
        return jsonify({"error": "Invalid OTP. Please try again."}), 400
@app.route("/dashboard")
def dashboard():
        return render_template("dashboard.html")
  # Otherwise, go to login page
@app.route("/result")
def result_page():
    # Query database for vote counts, sorting in descending order
    vote_counts = db.session.query(
        VotingRecord.candidate, db.func.count(VotingRecord.candidate)
    ).group_by(VotingRecord.candidate).order_by(db.func.count(VotingRecord.candidate).desc()).a
ll()
```

```
    # Convert results to a dictionary (sorted by votes)
    results = {candidate: count for candidate, count in vote_counts}

    # Identify the leading candidate
    leading_candidate = vote_counts[0] if vote_counts else None

    return render_template("result.html", results=results, leading_candidate=leading_candidate)
@app.route('/admins')
def admins_page():
    users = User.query.all()  # Assuming you have a User model in your database
    return render_template("admins.html", users=users)
@app.route('/video_records/<filename>')
def get_video(filename):
    return send_from_directory("static/video_records", filename)
if __name__ == "__main__":
    with app.app_context():
        db.create_all()
    app.run(debug=True)
```

HOME PAGE:

Dashboard.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Dashboard</title>
    <style>
     /* Navbar */
.navbar {
    display: flex;
    justify-content: space-between;
    align-items: center;
    background: #111;
    padding: 15px 40px;
    box-shadow: 0px 4px 10px rgba(0, 255, 0, 0.5);
    position: fixed;
```

```
    width: 100%;
    top: 0;
    left: 0;
    z-index: 1000;
     overflow: visible !important;
       width: 100vw;
    overflow: visible;

}

.navbar .logo img {
    width: 100px; /* Adjust logo size */
}

.navbar .menu {
    list-style: none;
    display: flex;
    gap: 30px;
        flex-wrap: wrap;
        margin-right:100px;



}

.navbar .menu li {
    display: inline;
        white-space: nowrap; /* Prevents text from breaking into multiple lines */

}


.navbar .menu li a {
    color: #00ff00;
    text-decoration: none;
    font-weight: bold;
    padding: 8px 12px;
```

```
    transition: 0.3s;

}

.navbar .menu li a:hover {
    background: #009900;
    border-radius: 5px;
}
    </style>

    <style>
        body {
            background-color: #111;
            color: #00ff00;
            font-family: Arial, sans-serif;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            margin: 0;
        }
        .container {
            background: #222;
            padding: 40px;
            border-radius: 10px;
            box-shadow: 0px 4px 10px rgba(0, 255, 0, 0.5);
            text-align: center;
            width: 80%;
            max-width: 800px;
            height: 300px; /* Increased height */
            display: flex;
            flex-direction: column;
            justify-content: center;
            margin-top:100px;
        }
        .register-btn {
```

```
            background: #00ff00;
            color: #111;
            border: none;
            padding: 12px 20px;
            font-size: 16px;
            font-weight: bold;
            cursor: pointer;
            border-radius: 5px;
            margin-top: 20px;
            text-decoration: none;
            display: inline-block;
        }
        .register-btn:hover {
            background: #009900;
        }
    </style>
</head>
<body>
<!-- Navbar -->
    <nav class="navbar">
        <div class="logo">
            <img src="{{ url_for('static', filename='party/logo.png') }}" alt="Logo">
        </div>
        <ul class="menu">
            <li><a href="{{ url_for('dashboard') }}">Home</a></li>
            <li><a href="{{ url_for('register') }}">Register</a></li>
            <li><a href="{{ url_for('login') }}">Login</a></li>
            <li><a href="{{ url_for('result_page') }}">Results</a></li>
            <li><a href="{{ url_for('admins_page') }}">Admin</a></li>
        </ul>
    </nav>
    <div class="container">
        <h1>Welcome to Blockchain-Based E-Voting System</h1>
       <a href="{{ url_for('register') }}" class="register-btn">User Register</a>
    </div>
</body>
```

```
</html>
```

REGISTER PAGE:

register.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Voter Registration</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='styles.css') }}">
    <style>
      body {
    font-family: Arial, sans-serif;
    background: black;
    display: flex;
    justify-content: center;
    align-items: center; /* Keeps form centered */
    min-height: 100vh;
    margin: 0;
    padding: 40px 0; /* Adds space around the form */
    margin-bottom:50px;
    padding-top:-50px;
}

.container {
    background: #111;
    padding: 20px;
    border-radius: 10px;
    box-shadow: 0px 4px 10px rgba(0, 255, 0, 0.5);
    width: 850px;
    max-width: 90%;
    text-align: center;
    color: white;
    margin-top: 750px; /* Moves form slightly down */
```

```
}

input, button {
    width: 90%;
    margin-bottom: 10px;
}

button {
    background: green;
    color: white;
    border: none;
    padding: 10px;
    border-radius: 5px;
    cursor: pointer;
}

/* Ensure button is always accessible */
button {
    bottom: 10px;
    background: green;
    color: white;
    border: none;
    padding: 10px;
    border-radius: 5px;
    cursor: pointer;
}
        h2 {
            color: #00ff00;
        }
        label {
            display: block;
            margin-top: 10px;
            font-weight: bold;
            color: #00ff00;
        }
        input {
```

```
        width: 95%;
        padding: 12px;
        font-size: 16px; /* Make text inside input more readable */
        margin-top: 5px;
        border: 1px solid #00ff00;
        border-radius: 5px;
        background: #222;
        color: white;
    }
    button {
        background: #00ff00;
        color: black;
        border: none;
        padding: 12px;
        margin-top: 10px;
        width: 100%;
        border-radius: 5px;
        cursor: pointer;
        font-weight: bold;
        font-size: 16px;
    }
    button:hover {
        background: #009900;
    }
    p {
        margin-top: 15px;
        color: #00ff00;
    }
    a {
        color: #00ff00;
        text-decoration: none;
        font-weight: bold;
    }
    /* Navbar */
.navbar {
  display: flex;
```

```
    justify-content: space-between;
    align-items: center;
    background: #111;
    padding: 15px 40px;
    box-shadow: 0px 4px 10px rgba(0, 255, 0, 0.5);
    position: fixed;
    width: 100%;
    top: 0;
    left: 0;
    z-index: 1000;
     overflow: visible !important;
       width: 100vw;
    overflow: visible;

}

.navbar .logo img {
    width: 100px; /* Adjust logo size */
}

.navbar .menu {
    list-style: none;
    display: flex;
    gap: 30px;
        flex-wrap: wrap;
        margin-right:100px;



}

.navbar .menu li {
    display: inline;
        white-space: nowrap; /* Prevents text from breaking into multiple lines */

}
```

```
.navbar .menu li a {
    color: #00ff00;
    text-decoration: none;
    font-weight: bold;
    padding: 8px 12px;
    transition: 0.3s;

}

.navbar .menu li a:hover {
    background: #009900;
    border-radius: 5px;
}

    </style>
    <script>
    function sendOTP() {
        let phone = document.getElementById("phoneNumber").value;

        if (!phone) {
            alert("Please enter a phone number.");
            return;
        }

        fetch("/send_otp", {
            method: "POST",
            headers: { "Content-Type": "application/x-www-form-urlencoded" },
            body: "phoneNumber=" + encodeURIComponent(phone)
        })
        .then(response => response.json())
        .then(data => {
            if (data.message) {
                alert(data.message);
            } else {
                alert("Error: " + data.error);
```

```
            }
        })
        .catch(error => console.error("Error:", error));
    }
</script>
<script>
    function verifyOTP() {
        let otp = document.getElementById("otp").value;
        let phone = document.getElementById("phoneNumber").value;

        fetch("/verify_otp", {
            method: "POST",
            headers: { "Content-Type": "application/json" },
            body: JSON.stringify({ phoneNumber: phone, otp: otp })
        })
        .then(response => response.json())
        .then(data => {
            if (data.message) {
                alert(data.message);  // ✅ Show success message
            } else {
                alert("Error: " + data.error);  // ❌ Show error message
            }
        })
        .catch(error => console.error("Error:", error));
    }
</script>
<script>
    let otpVerified = false;

    function sendOTP() {
        let phone = document.getElementById("phoneNumber").value;

        if (!phone) {
            alert("Please enter a phone number.");
            return;
        }
```

```
    fetch("/send_otp", {
        method: "POST",
        headers: { "Content-Type": "application/x-www-form-urlencoded" },
        body: "phoneNumber=" + encodeURIComponent(phone)
    })
    .then(response => response.json())
    .then(data => {
        alert(data.message);
    })
    .catch(error => console.error("Error:", error));
}

function verifyOTP() {
    let otp = document.getElementById("otp").value;
    let phone = document.getElementById("phoneNumber").value;

    fetch("/verify_otp", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify({ phoneNumber: phone, otp: otp })
    })
    .then(response => response.json())
    .then(data => {
        if (data.message) {
            alert(data.message);
            otpVerified = true;
            document.getElementById("registerBtn").disabled = false;  // Enable Register button
        } else {
            alert("Error: " + data.error);
            otpVerified = false;
            document.getElementById("registerBtn").disabled = true;  // Keep Register button
disabled
        }
    })
    .catch(error => console.error("Error:", error));
```

```
    }

    document.getElementById("registerForm").addEventListener("submit", function(event) {
        if (!otpVerified) {
            event.preventDefault();  // Stop form submission
            alert("Please enter a valid OTP before registering.");
        }
    });
</script>

<script>
    let otpVerified = false;

    function sendOTP() {
        let phone = document.getElementById("phoneNumber").value;

        if (!phone) {
            alert("Please enter a phone number.");
            return;
        }

        fetch("/send_otp", {
            method: "POST",
            headers: { "Content-Type": "application/x-www-form-urlencoded" },
            body: "phoneNumber=" + encodeURIComponent(phone)
        })
        .then(response => response.json())
        .then(data => {
            alert(data.message);
        })
        .catch(error => console.error("Error:", error));
    }

    function verifyOTP() {
        let otp = document.getElementById("otp").value;
        let phone = document.getElementById("phoneNumber").value;
```

```
        fetch("/verify_otp", {
            method: "POST",
            headers: { "Content-Type": "application/json" },
            body: JSON.stringify({ phoneNumber: phone, otp: otp })
        })
        .then(response => response.json())
        .then(data => {
            if (data.message) {
                alert(data.message);
                otpVerified = true;
                document.getElementById("registerBtn").disabled = false;  // Enable Register button
            } else {
                alert("Error: " + data.error);
                otpVerified = false;
                document.getElementById("registerBtn").disabled = true;  // Keep Register button
disabled
            }
        })
        .catch(error => console.error("Error:", error));
    }
</script>
<script>
document.getElementById("first_name").addEventListener("input", function () {
    this.value = this.value.replace(/[^A-Za-z]/g, '');
});

document.getElementById("last_name").addEventListener("input", function () {
    this.value = this.value.replace(/[^A-Za-z]/g, '');
});

document.getElementById("aadhaar_id").addEventListener("input", function () {
    this.value = this.value.replace(/[^0-9]/g, '');
    if (this.value.length > 12) {
        alert("Aadhaar ID must be exactly 12 digits.");
        this.value = this.value.slice(0, 12);
```

```
    }
});

document.getElementById("voter_id").addEventListener("input", function () {
    this.value = this.value.replace(/[^A-Za-z0-9]/g, '');
    if (this.value.length > 10) {
        alert("Voter ID must be exactly 10 characters (letters and digits).");
        this.value = this.value.slice(0, 10);
    }
});
</script>



</head>
<!-- Navbar -->
    <nav class="navbar">
        <div class="logo">
            <img src="{{ url_for('static', filename='party/logo.png') }}" alt="Logo">
        </div>
        <ul class="menu">
            <li><a href="{{ url_for('dashboard') }}">Home</a></li>
            <li><a href="{{ url_for('register') }}">Register</a></li>
            <li><a href="{{ url_for('login') }}">Login</a></li>
            <li><a href="{{ url_for('result_page') }}">Results</a></li>
            <li><a href="{{ url_for('admins_page') }}">Admin</a></li>
        </ul>
    </nav>
<body>
    <div class="container">
        <h2>Voter Registration</h2>
        <form id="registerForm" action="/" method="POST" enctype="multipart/form-data">
         <label for="first_name">First Name:</label>
    <input type="text" id="first_name" name="first_name" required>
    <label for="last_name">Last Name:</label>
    <input type="text" id="last_name" name="last_name" required>
    <label for="voterId">Voter ID:</label>
```

```
    <input type="text" id="voterId" name="voterId" required maxlength="10">
    <label for="aadhaar_id">Aadhaar ID:</label>
    <input type="text" id="aadhaar_id" name="aadhaar_id" required maxlength="12">

    <label for="phoneNumber">Phone Number:</label>
    <input type="text" id="phoneNumber" name="phoneNumber" required>
    <button type="button" onclick="sendOTP()">Send OTP</button>

    <label for="otp">Enter OTP:</label>
    <input type="text" id="otp" name="otp" required>
    <button type="button" onclick="verifyOTP()">Verify OTP</button>

    <label for="aadhaarCard">Upload Aadhaar Card:</label>
    <input type="file" id="aadhaarCard" name="aadhaarCard" accept="image/*" required>

    <button type="submit" id="registerBtn" disabled>Register</button>
</form>

        <p>Already registered? <a href="{{ url_for('login') }}">Login here</a></p>
        <a href="{{ url_for('result_page') }}" class="btn btn-primary">View Results</a>
        <a href="{{ url_for('admins_page') }}" class="admin-link">Admin Panel</a>


    </div>


</body>
</html>
```

LOGIN PAGE:

login.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Voter Login</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='styles.css') }}">
    <style>
    /* Global Styles */
    body {
        font-family: Arial, sans-serif;
        background-color: black; /* Full black background */
        margin: 0;
        padding: 0;
        display: flex;
        justify-content: center;
        align-items: center;
        height: 100vh;
        color: white;
        flex-direction: column;
        margin-top:200px;
        margin-bottom:50px;
    }

    .container {
        background: #111;
        padding: 20px;
        border-radius: 10px;
        box-shadow: 0px 4px 10px rgba(0, 255, 0, 0.5);
        max-width: 800px; /* Increased width for better alignment */
        width: 100%;
        text-align: center;
    }
```

```
h2, h1 {
    color: #00ff00; /* Green for headings */
}

label {
    display: block;
    margin-top: 10px;
    font-weight: bold;
    color: #00ff00;
}

input, select {
    width: 100%;
    padding: 8px;
    margin-top: 5px;
    border: 1px solid #00ff00;
    border-radius: 5px;
    background: #222;
    color: white;
}

button {
    background: #00ff00;
    color: black;
    border: none;
    padding: 10px;
    margin-top: 10px;
    width: 100%;
    border-radius: 5px;
    cursor: pointer;
    font-weight: bold;
}

button:hover {
    background: #009900;
}
```

```
p {
    margin-top: 15px;
    color: #00ff00;
}

a {
    color: #00ff00;
    text-decoration: none;
    font-weight: bold;
}

a:hover {
    color: #009900;
}

/* Admin Panel Specific Styles */
h1 {
    text-align: center;
    margin-bottom: 40px;
}

form {
    display: flex;
    flex-direction: column;
    align-items: center;
}

form label {
    text-align: left;
    width: 100%;
    margin-bottom: 5px;
}

form input[type="text"], form input[type="file"] {
    width: 80%;
```

```
}

form button {
    width: 100%;
    margin-top: 20px;
}

/* Table Styles */
table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 20px;
}

table th, table td {
    padding: 10px;
    text-align: center;
    border: 1px solid #444;
}

table th {
    background-color: #333;
}

table tr:nth-child(even) {
    background-color: #444;
}

table img {
    border-radius: 5px;
}

/* Timer and Video Styling for Voting Page */
#timer {
    font-size: 20px;
    margin-top: 10px;
```

```
        color: #00ff00;
    }

    #video {
        margin-top: 20px;
        width: 100%;
        max-width: 400px;
        height: auto;
    }

    #canvas {
        display: none;
    }
</style>
    <style>
     /* Navbar */
.navbar {
    display: flex;
    justify-content: space-between;
    align-items: center;
    background: #111;
    padding: 15px 40px;
    box-shadow: 0px 4px 10px rgba(0, 255, 0, 0.5);
    position: fixed;
    width: 100%;
    top: 0;
    left: 0;
    z-index: 1000;
     overflow: visible !important;
       width: 100vw;
    overflow: visible;

}

.navbar .logo img {
    width: 100px; /* Adjust logo size */
```

```
}

.navbar .menu {
    list-style: none;
    display: flex;
    gap: 30px;
        flex-wrap: wrap;
        margin-right:100px;



}

.navbar .menu li {
    display: inline;
        white-space: nowrap; /* Prevents text from breaking into multiple lines */

}


.navbar .menu li a {
    color: #00ff00;
    text-decoration: none;
    font-weight: bold;
    padding: 8px 12px;
    transition: 0.3s;

}

.navbar .menu li a:hover {
    background: #009900;
    border-radius: 5px;
}
    </style>

    <script>
        let otpVerified = false;
```

```
    function sendLoginOTP() {
        let phone = document.getElementById("phoneNumber").value;
        let voterId = document.getElementById("voterId").value;

        if (!phone || !voterId) {
            alert("Please enter Voter ID and Phone Number.");
            return;
        }

        fetch("/send_login_otp", {
            method: "POST",
            headers: { "Content-Type": "application/x-www-form-urlencoded" },
            body: "voterId=" + encodeURIComponent(voterId) + "&phoneNumber=" +
encodeURIComponent(phone)
        })
        .then(response => response.json())
        .then(data => {
            if (data.message) {
                alert(data.message);  // ✅ Show success message
            } else {
                alert("Error: " + data.error);  // ❌ Show error message
            }
        })
        .catch(error => console.error("Error:", error));
    }

    function verify_login_otp() {
        let otp = document.getElementById("otp").value;
        let phone = document.getElementById("phoneNumber").value;

        if (!otp || !phone) {
            alert("Please enter OTP and phone number.");
            return;
        }
```

```
        fetch("/verify_login_otp", {
            method: "POST",
            headers: { "Content-Type": "application/json" },
            body: JSON.stringify({ phoneNumber: phone, otp: otp })
        })
        .then(response => response.json())
        .then(data => {
            if (data.message) {
                alert(data.message);  // ✅ Show success message
                otpVerified = true;
                document.getElementById("loginBtn").disabled = false;  // Enable Login button
            } else {
                alert("Error: " + data.error);
                otpVerified = false;
                document.getElementById("loginBtn").disabled = true;  // Keep Login button
disabled
            }
        })
        .catch(error => console.error("Error:", error));
    }

    function redirectToVotingPage(event) {
        if (!otpVerified) {
            event.preventDefault();  // ❌ Stop form submission if OTP is not verified
            alert("Please verify OTP before logging in.");
        } else {
            window.location.href = "/voting_page";  // ✅ Redirect to voting page
        }
    }
</script>
<script>
document.getElementById("loginForm").addEventListener("submit", function(event) {
    event.preventDefault();

    fetch("/login", {
        method: "POST",
```

```
            body: new FormData(this)
        }).then(response => response.text())
          .then(data => {
            if (data.includes("Invalid OTP")) {
                alert("Incorrect OTP! Try again.");
            } else {
                window.location.href = "/voting_page"; // Redirect to voting page
            }
          });
    });
</script>


</head>
<!-- Navbar -->
    <nav class="navbar">
        <div class="logo">
            <img src="{{ url_for('static', filename='party/logo.png') }}" alt="Logo">
        </div>
        <ul class="menu">
            <li><a href="/">Home</a></li>
            <li><a href="{{ url_for('register') }}">Register</a></li>
            <li><a href="{{ url_for('login') }}">Login</a></li>
            <li><a href="{{ url_for('result_page') }}">Results</a></li>
            <li><a href="{{ url_for('admins_page') }}">Admin</a></li>
        </ul>
    </nav>
<body>
    <div class="container">
        <h2>Voter Login</h2>
<form id="loginForm" action="{{ url_for('login') }}" method="POST">
            <label for="voterId">Voter ID:</label>
            <input type="text" id="voterId" name="voterId" required>

            <label for="phoneNumber">Phone Number:</label>
            <input type="text" id="phoneNumber" name="phoneNumber" required>
```

```
        <button type="button" onclick="sendLoginOTP()">Send OTP</button>

        <label for="otp">Enter OTP:</label>
        <input type="text" id="otp" name="otp" required>
        <button type="button" onclick="verify_login_otp()">Verify OTP</button>

        <!-- ✅ Ensure the login button redirects to voting page only after OTP verification -->
        <button type="submit" id="loginBtn" disabled>Login</button>

      </form>
      <p>Not registered yet? <a href="{{ url_for('register') }}">Register here</a></p>
    </div>

</body>
</html>
```

VOTING PAGE:

Voting_page.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Voting Page</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='styles.css') }}">
    <script src="https://cdnjs.cloudflare.com/ajax/libs/Chart.js/3.9.1/chart.min.js"></script>
     <style>
     /* Global Styles */
     body {
        font-family: Arial, sans-serif;
        background-color: black; /* Full black background */
        margin: 0;
        padding: 0;
        display: flex;
        justify-content: center;
        align-items: center;
        height: 100vh;
```

```
    color: white;
    flex-direction: column;
    margin-top:200px;
    margin-bottom:50px;
}

.container {
    background: #111;
    padding: 20px;
    border-radius: 10px;
    box-shadow: 0px 4px 10px rgba(0, 255, 0, 0.5);
    max-width: 800px; /* Increased width for better alignment */
    width: 100%;
    text-align: center;
}

h2, h1 {
    color: #00ff00; /* Green for headings */
}

label {
    display: block;
    margin-top: 10px;
    font-weight: bold;
    color: #00ff00;
}

input, select {
    width: 100%;
    padding: 8px;
    margin-top: 5px;
    border: 1px solid #00ff00;
    border-radius: 5px;
    background: #222;
    color: white;
}
```

```
button {
    background: #00ff00;
    color: black;
    border: none;
    padding: 10px;
    margin-top: 10px;
    width: 100%;
    border-radius: 5px;
    cursor: pointer;
    font-weight: bold;
}

button:hover {
    background: #009900;
}

p {
    margin-top: 15px;
    color: #00ff00;
}

a {
    color: #00ff00;
    text-decoration: none;
    font-weight: bold;
}

a:hover {
    color: #009900;
}

/* Admin Panel Specific Styles */
h1 {
    text-align: center;
    margin-bottom: 40px;
```

```
}

form {
    display: flex;
    flex-direction: column;
    align-items: center;
}

form label {
    text-align: left;
    width: 100%;
    margin-bottom: 5px;
}

form input[type="text"], form input[type="file"] {
    width: 80%;
}

form button {
    width: 100%;
    margin-top: 20px;
}

/* Table Styles */
table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 20px;
}

table th, table td {
    padding: 10px;
    text-align: center;
    border: 1px solid #444;
}
```

```
    table th {
        background-color: #333;
    }

    table tr:nth-child(even) {
        background-color: #444;
    }

    table img {
        border-radius: 5px;
    }

    /* Timer and Video Styling for Voting Page */
    #timer {
        font-size: 20px;
        margin-top: 10px;
        color: #00ff00;
    }

    #video {
        margin-top: 20px;
        width: 100%;
        max-width: 400px;
        height: auto;
    }

    #canvas {
        display: none;
    }
</style>
  <style>
   /* Navbar */
.navbar {
  display: flex;
  justify-content: space-between;
  align-items: center;
```

```
    background: #111;
    padding: 15px 40px;
    box-shadow: 0px 4px 10px rgba(0, 255, 0, 0.5);
    position: fixed;
    width: 100%;
    top: 0;
    left: 0;
    z-index: 1000;
     overflow: visible !important;
       width: 100vw;
    overflow: visible;

}

.navbar .logo img {
    width: 100px; /* Adjust logo size */
}

.navbar .menu {
    list-style: none;
    display: flex;
    gap: 30px;
        flex-wrap: wrap;
        margin-right:100px;


}

.navbar .menu li {
    display: inline;
        white-space: nowrap; /* Prevents text from breaking into multiple lines */

}


.navbar .menu li a {
```

```
    color: #00ff00;
    text-decoration: none;
    font-weight: bold;
    padding: 8px 12px;
    transition: 0.3s;

}

.navbar .menu li a:hover {
    background: #009900;
    border-radius: 5px;
}
    </style>
</head>
<!-- Navbar -->
    <nav class="navbar">
        <div class="logo">
            <img src="{{ url_for('static', filename='party/logo.png') }}" alt="Logo">
        </div>
        <ul class="menu">
            <li><a href="/">Home</a></li>
            <li><a href="{{ url_for('register') }}">Register</a></li>
            <li><a href="{{ url_for('login') }}">Login</a></li>
            <li><a href="{{ url_for('result_page') }}">Results</a></li>
            <li><a href="{{ url_for('admins_page') }}">Admin</a></li>
        </ul>
    </nav>
<body>
    <div class="container">
        <h2>Cast Your Vote</h2>
        <p id="timer">Time Left: 5:00</p>
        <video id="video" width="400" height="300" autoplay></video>

        {% if expiry_time %}
            <p>Voting session expires in: {{ expiry_time - time.time() }} seconds</p>
        {% endif %}
```

```
      <form id="voteForm" action="/submit_vote" method="POST" enctype="multipart/form-data">
          <label>Select Your Party:</label>
          <select id="candidate" name="candidate" required>
            <option value="AIADMK(All India Anna Dravida Munnetra Kazhagam)">AIADMK(All India Anna Dravida Munnetra Kazhagam)</option>
            <option value="DMK(Dravida Munnetra Kazhagam)">DMK(Dravida Munnetra Kazhagam)</option>
            <option value="CPI(Communist Party of India)">CPI(Communist Party of India)</option>
            <option value="DMDK(Desiya Murpokku Dravida Kazhagam)">DMDK(Desiya Murpokku Dravida Kazhagam)</option>
            <option value="NTK(Naam Tamilar Katchi)">NTK(Naam Tamilar Katchi)</option>
            <option value="VCK(Viduthalai Chiruthaigal Katchi)">VCK(Viduthalai Chiruthaigal Katchi)</option>
            <option value="AAP(Aam Aadmi Party)">AAP(Aam Aadmi Party)</option>
            <option value="BSP(Bahujan Samaj Party)">BSP(Bahujan Samaj Party)</option>
            <option value="BJP(Bharatiya Janata Party)">BJP(Bharatiya Janata Party)</option>
            <option value="INC(Indian National Congress)">INC(Indian National Congress)</option>
          </select>
          <button type="submit">Submit Vote</button>
      </form>
    </div>

    <script>
    navigator.mediaDevices.getUserMedia({ video: true, audio: true })
   .then(stream => {
    let mediaRecorder = new MediaRecorder(stream);
    let chunks = [];

    mediaRecorder.ondataavailable = event => {
      chunks.push(event.data);
    };
```

```
  mediaRecorder.onstop = () => {
    let blob = new Blob(chunks, { type: 'video/mp4' });
    let formData = new FormData();

    let voterID = document.getElementById("voter_id").value;  // Get voter ID from form
    let timestamp = Date.now();
    let videoFilename = `${voterID}_${timestamp}.mp4`; // Unique filename

    formData.append("video", blob, videoFilename);  // Send video with filename
    formData.append("voter_id", voterID);  // Also send voter ID

    fetch("/submit_vote", {
      method: "POST",
      body: formData
    })
    .then(response => response.json())
    .then(data => {
      alert(data.message);
    })
    .catch(error => {
      console.error("Error:", error);
    });
  };

  mediaRecorder.start();

  setTimeout(() => {
    mediaRecorder.stop();
  }, 5000); // Record for 5 seconds
})
.catch(error => {
  console.error("Error accessing webcam:", error);
});
</script>

  <script>
```

```
    let timeLeft = {% if expiry_time %} {{ expiry_time - time.time() }} {% else %} 300 {% endif %}; // Default 5 minutes

    function updateTimer() {
        let minutes = Math.floor(timeLeft / 60);
        let seconds = Math.floor(timeLeft % 60);
        document.getElementById("timer").textContent = `Time Left: ${minutes}:${seconds < 10 ? "0" : ""}${seconds}`;

        if (timeLeft <= 0) {
            clearInterval(interval);
            alert("Your time has ended. Redirecting...");
            window.location.href = "/logout";
        } else {
            timeLeft--;
        }
    }
    let interval = setInterval(updateTimer, 1000);
</script>

<script>
    let video = document.getElementById("video");
    let mediaRecorder;
    let recordedChunks = [];

    navigator.mediaDevices.getUserMedia({ video: true, audio: true })
        .then(stream => {
            video.srcObject = stream;
            mediaRecorder = new MediaRecorder(stream);
            mediaRecorder.ondataavailable = event => recordedChunks.push(event.data);
            mediaRecorder.start();
        })
        .catch(err => {
            console.error("Camera access denied:", err);
            alert("Camera access is required for voting.");
        });
```

```
        // Stop recording & submit form
        document.getElementById("voteForm").addEventListener("submit", function (event) {
            event.preventDefault(); // Stop form submission
            if (mediaRecorder) mediaRecorder.stop();
            mediaRecorder.onstop = () => {
                let blob = new Blob(recordedChunks, { type: "video/webm" });
                let formData = new FormData(this);
                formData.append("video", blob, "recording.webm");

                fetch("/submit_vote", { method: "POST", body: formData })
                    .then(response => response.text())
                    .then(data => {
                        alert("Vote submitted successfully!");
                        window.location.href = "/logout";
                    })
                    .catch(error => console.error("Error submitting vote:", error));
            };
        });
    </script>
</body>
</html>
```

BLOCKCHAIN:

Blockchain.py

```
import hashlib
import json
import time
from flask import Flask, request, jsonify
import sqlite3

class Blockchain:
    def __init__(self):
        self.chain = []
        self.current_votes = []
        self.create_block(previous_hash='1', proof=100)  # Genesis block
```

```
def create_block(self, proof, previous_hash):
    block = {
        'index': len(self.chain) + 1,
        'timestamp': time.time(),
        'votes': self.current_votes,
        'proof': proof,
        'previous_hash': previous_hash
    }
    self.current_votes = []
    self.chain.append(block)
    return block

def add_vote(self, voter_id, candidate):
    vote = {
        'voter_id': voter_id,
        'candidate': candidate
    }
    self.current_votes.append(vote)
    return self.get_last_block()['index'] + 1

def get_last_block(self):
    return self.chain[-1]

def proof_of_work(self, previous_proof):
    new_proof = 1
    while not self.valid_proof(new_proof, previous_proof):
        new_proof += 1
    return new_proof

def valid_proof(self, new_proof, previous_proof):
    guess = f'{new_proof}{previous_proof}'.encode()
    guess_hash = hashlib.sha256(guess).hexdigest()
    return guess_hash[:4] == "0000"

def hash(self, block):
    encoded_block = json.dumps(block, sort_keys=True).encode()
```

```
        return hashlib.sha256(encoded_block).hexdigest()

app = Flask(__name__)
blockchain = Blockchain()

def store_vote_in_db(voter_id, candidate):
    conn = sqlite3.connect('votes.db')
    cursor = conn.cursor()
    cursor.execute("INSERT INTO votes (voter_id, candidate) VALUES (?, ?)", (voter_id,
candidate))
    conn.commit()
    conn.close()

@app.route('/vote', methods=['POST'])
def vote():
    data = request.get_json()
    voter_id = data['voter_id']
    candidate = data['candidate']

    blockchain.add_vote(voter_id, candidate)
    store_vote_in_db(voter_id, candidate)
    print(json.dumps(blockchain.chain, indent=4))



    return jsonify({'message': 'Vote has been cast successfully'}), 201

@app.route('/get_chain', methods=['GET'])
def get_chain():
    response = {'chain': blockchain.chain, 'length': len(blockchain.chain)}
    return jsonify(response), 200

@app.route('/mine', methods=['GET'])
def mine_block():
    previous_block = blockchain.get_last_block()
    previous_proof = previous_block['proof']
    proof = blockchain.proof_of_work(previous_proof)
```

```
    previous_hash = blockchain.hash(previous_block)
    block = blockchain.create_block(proof, previous_hash)
    return jsonify({'message': 'New block mined!', 'block': block}), 200

if __name__ == '__main__':
    app.run(debug=True)
```

ADMIN PAGE:

admins.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Admin Panel - User Details</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
    <style>
     /* Navbar */
.navbar {
    display: flex;
    justify-content: space-between;
    align-items: center;
    background: #111;
    padding: 15px 40px;
    box-shadow: 0px 4px 10px rgba(0, 255, 0, 0.5);
    position: fixed;
    width: 100%;
    top: 0;
    left: 0;
    z-index: 1000;
     overflow: visible !important;
       width: 100vw;
    overflow: visible;

}

.navbar .logo img {
```

```
    width: 100px; /* Adjust logo size */
}

.navbar .menu {
    list-style: none;
    display: flex;
    gap: 30px;
        flex-wrap: wrap;
        margin-right:100px;



}

.navbar .menu li {
    display: inline;
        white-space: nowrap; /* Prevents text from breaking into multiple lines */

}


.navbar .menu li a {
    color: #00ff00;
    text-decoration: none;
    font-weight: bold;
    padding: 8px 12px;
    transition: 0.3s;

}

.navbar .menu li a:hover {
    background: #009900;
    border-radius: 5px;
}</style>
    <script>
    setInterval(function() {
        location.reload();
```

```
    }, 500000); // Refresh every 5 seconds
</script>

    <style>
        body {
            background-color: black;
            color: white;
            font-family: Arial, sans-serif;
            text-align: center;
            margin-top:200px;
            margin-bottom:20px;
        }
        h1 {
            color: limegreen;
            margin-top: 20px;
        }
        table {
            width: 90%;
            margin: 20px auto;
            border-collapse: collapse;
            background-color: #222;
            box-shadow: 0 0 10px limegreen;
        }
        th, td {
            border: 1px solid limegreen;
            padding: 10px;
            text-align: center;
        }
        th {
            background-color: limegreen;
            color: black;
            font-weight: bold;
        }
        tr:nth-child(even) {
            background-color: #333;
        }
```

```
        tr:hover {
            background-color: limegreen;
            color: black;
            transition: 0.3s;
        }
        .btn {
            display: inline-block;
            padding: 10px 20px;
            margin-top: 10px;
            color: black;
            background-color: limegreen;
            border: none;
            cursor: pointer;
            text-decoration: none;
            font-weight: bold;
        }
        .btn:hover {
            background-color: white;
            color: black;
        }
    </style>
</head>
<!-- Navbar -->
    <nav class="navbar">
        <div class="logo">
            <img src="{{ url_for('static', filename='party/logo.png') }}" alt="Logo">
        </div>
        <ul class="menu">
            <li><a href="{{ url_for('dashboard') }}">Home</a></li>
            <li><a href="{{ url_for('register') }}">Register</a></li>
            <li><a href="{{ url_for('login') }}">Login</a></li>
            <li><a href="{{ url_for('result_page') }}">Results</a></li>
            <li><a href="{{ url_for('admins_page') }}">Admin</a></li>
        </ul>
    </nav>
<body>
```

```
<h1>Admin Panel - Registered Users</h1>

<table>
  <tr>
    <th>ID</th>
    <th>First Name</th>
    <th>Last Name</th>
    <th>Voter ID</th>
    <th>Aadhaar ID</th>
    <th>Phone Number</th>
    <th>Voted</th>
    <th>Aadhaar Card</th>
    <th>Video</th>
  </tr>
  {% for user in users %}
  <tr>
    <td>{{ user.id }}</td>
    <td>{{ user.first_name }}</td>
    <td>{{ user.last_name }}</td>
    <td>{{ user.voter_id }}</td>
    <td>{{ user.aadhaar_id }}</td>
    <td>{{ user.phone_number }}</td>
    <td>{{ "Yes" if user.voted else "No" }}</td>
    <td>
    {% if user.aadhaar_filename %}
  <td><a href="{{ user.aadhaar_card_path }}" target="_blank">View Aadhaar</a></td>
    {% else %}
      No Aadhaar Uploaded
    {% endif %}
    </td>
    <td>
  {% if user.video_filename %}
    <a href="{{ url_for('static', filename='video_records/' + user.video_filename) }}"
target="_blank">Watch Video</a>
  {% else %}
```

```
        No Video Uploaded
    {% endif %}
</td>
      </tr>
       {% endfor %}
    </table>

    <a href="/" class="btn">Back to Home</a>

</body>
</html>
```

RESULT PAGE:

result.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Election Results</title>
    <style>
     /* Navbar */
.navbar {
    display: flex;
    justify-content: space-between;
    align-items: center;
    background: #111;
    padding: 15px 40px;
    box-shadow: 0px 4px 10px rgba(0, 255, 0, 0.5);
    position: fixed;
    width: 100%;
    top: 0;
    left: 0;
    z-index: 1000;
     overflow: visible !important;
       width: 100vw;
    overflow: visible;
```

```
}

.navbar .logo img {
    width: 100px; /* Adjust logo size */
}

.navbar .menu {
    list-style: none;
    display: flex;
    gap: 30px;
        flex-wrap: wrap;
        margin-right:100px;


}

.navbar .menu li {
    display: inline;
        white-space: nowrap; /* Prevents text from breaking into multiple lines */

}


.navbar .menu li a {
    color: #00ff00;
    text-decoration: none;
    font-weight: bold;
    padding: 8px 12px;
    transition: 0.3s;

}

.navbar .menu li a:hover {
    background: #009900;
    border-radius: 5px;
```

```
}</style>
  <style>
    body {
      background-color: black;
      color: green;
      font-family: Arial, sans-serif;
      text-align: center;
      margin-top:200px;
      margin-bottom:50px;
    }
    h1 {
      color: limegreen;
    }
    table {
      width: 50%;
      margin: 20px auto;
      border-collapse: collapse;
      background-color: #1a1a1a;
    }
    th, td {
      border: 1px solid green;
      padding: 10px;
      text-align: center;
      color:white;
    }
    th {
      background-color: darkgreen;
      color:white;
    }
    .lead {
      font-size: 1.2em;
      color: limegreen;
      margin-top: 20px;
    }
    .voter-choice {
      font-size: 1.1em;
```

```
            color: white;
            margin-top: 10px;
        }
    </style>
</head>
<!-- Navbar -->
    <nav class="navbar">
        <div class="logo">
            <img src="{{ url_for('static', filename='party/logo.png') }}" alt="Logo">
        </div>
        <ul class="menu">
            <li><a href="/">Home</a></li>
            <li><a href="{{ url_for('register') }}">Register</a></li>
            <li><a href="{{ url_for('login') }}">Login</a></li>
            <li><a href="{{ url_for('result_page') }}">Results</a></li>
            <li><a href="{{ url_for('admins_page') }}">Admin</a></li>
        </ul>
    </nav>
<body>
    <h1>Election Results</h1>

    <!-- Voter's Choice -->
    {% if voter_choice %}
    <div class="voter-choice">
        {{ voter_name }} ({{ voter_id }}) voted for {{ voter_choice }}
    </div>
    {% endif %}

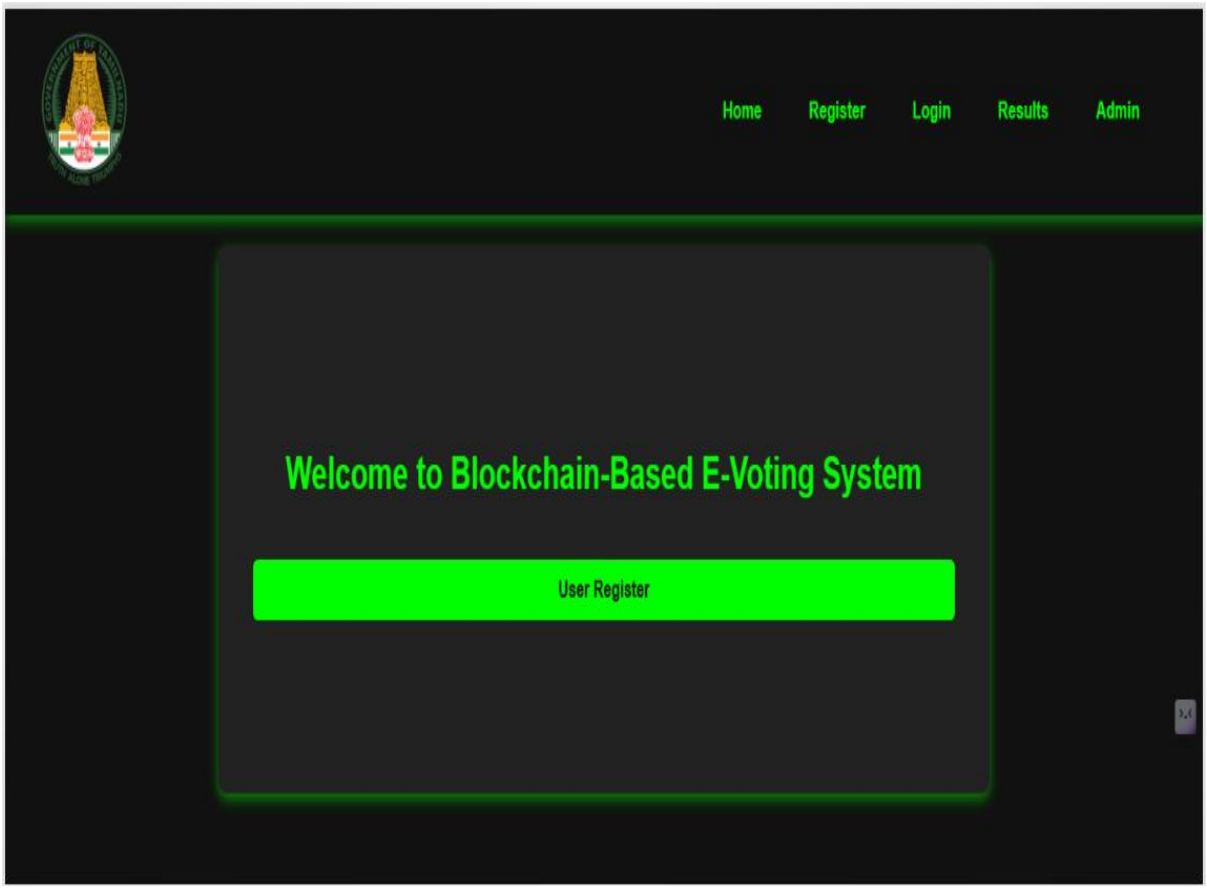
    <!-- Table displaying vote counts and percentages -->
   <table border="1">
    <tr>
        <th>parties</th>
        <th>Vote Count</th>
    </tr>
    {% for candidate, count in results.items() %}
    <tr>
```

```
        <td>{{ candidate }}</td>
        <td>{{ count }}</td>
    </tr>
    {% endfor %}
</table>

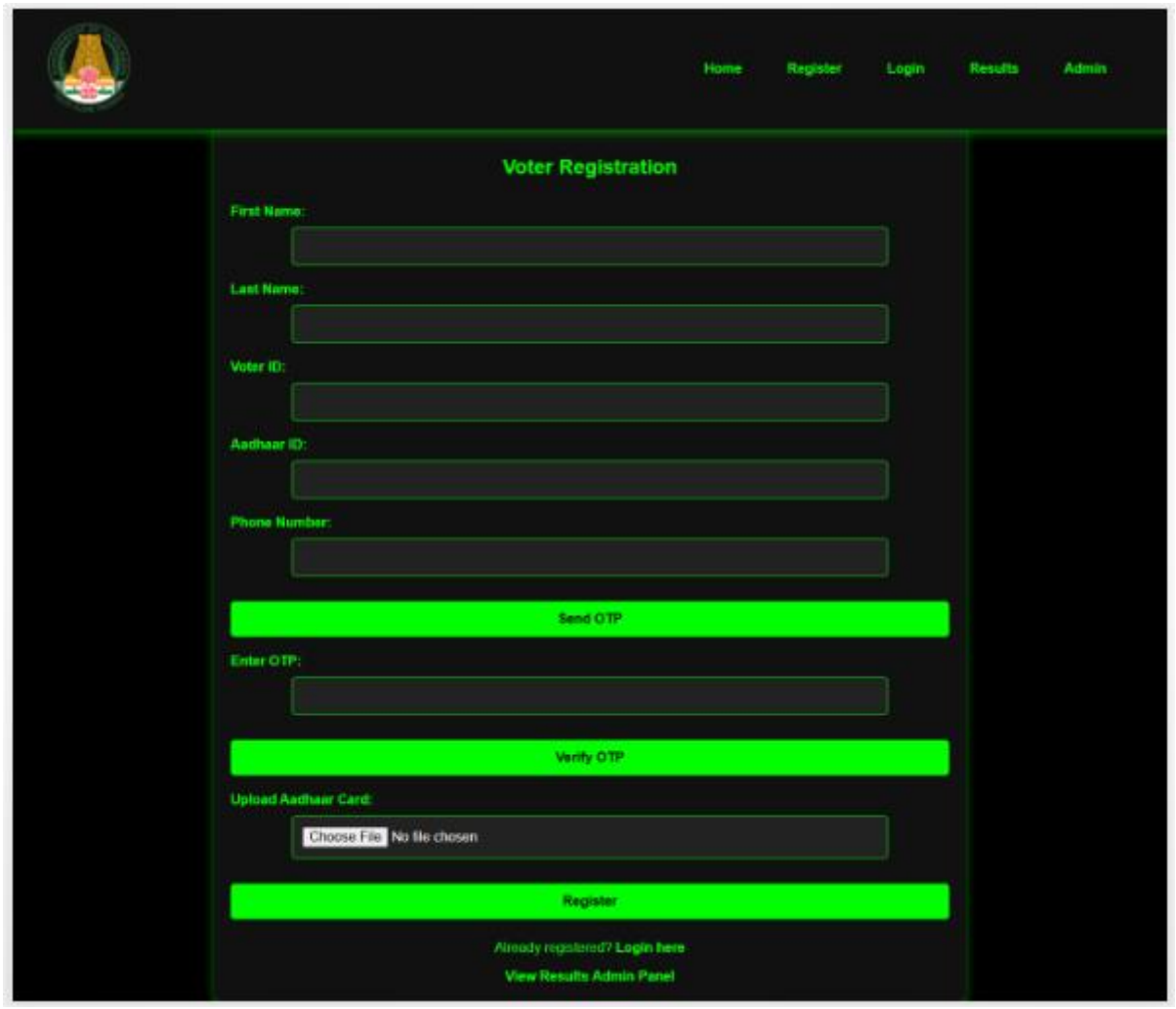
{% if leading_candidate %}
<h2>    Leading party: {{ leading_candidate[0] }} with {{ leading_candidate[1] }} votes!</h2>
{% else %}
<h2>No votes have been cast yet.</h2>
{% endif %}
</body>
</html>
```

SCREEN SHOTS:

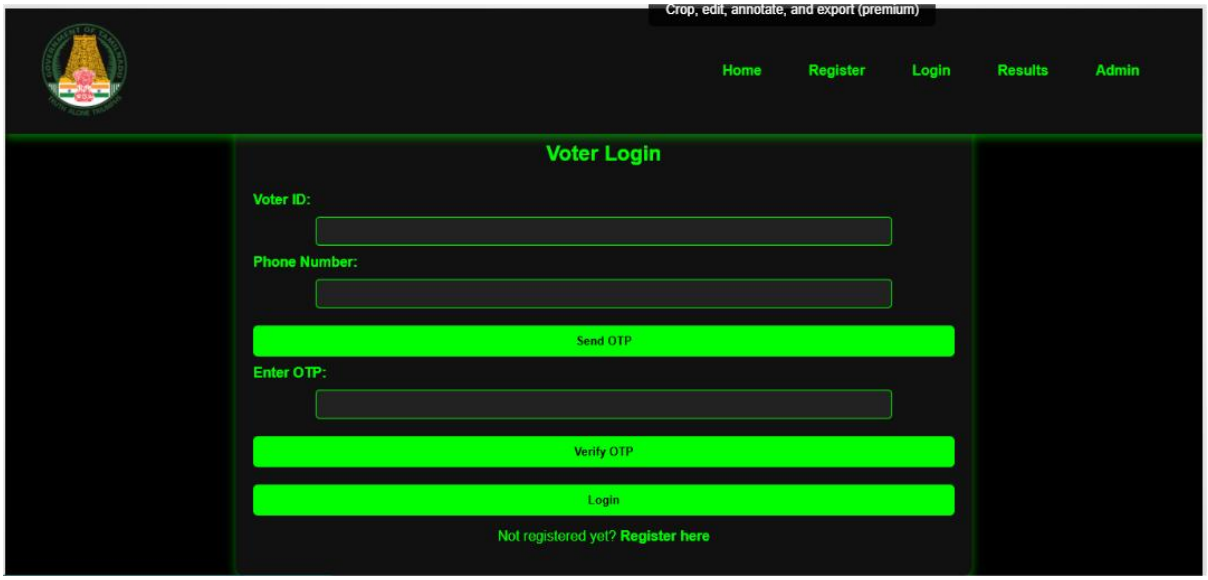
HOME PAGE:



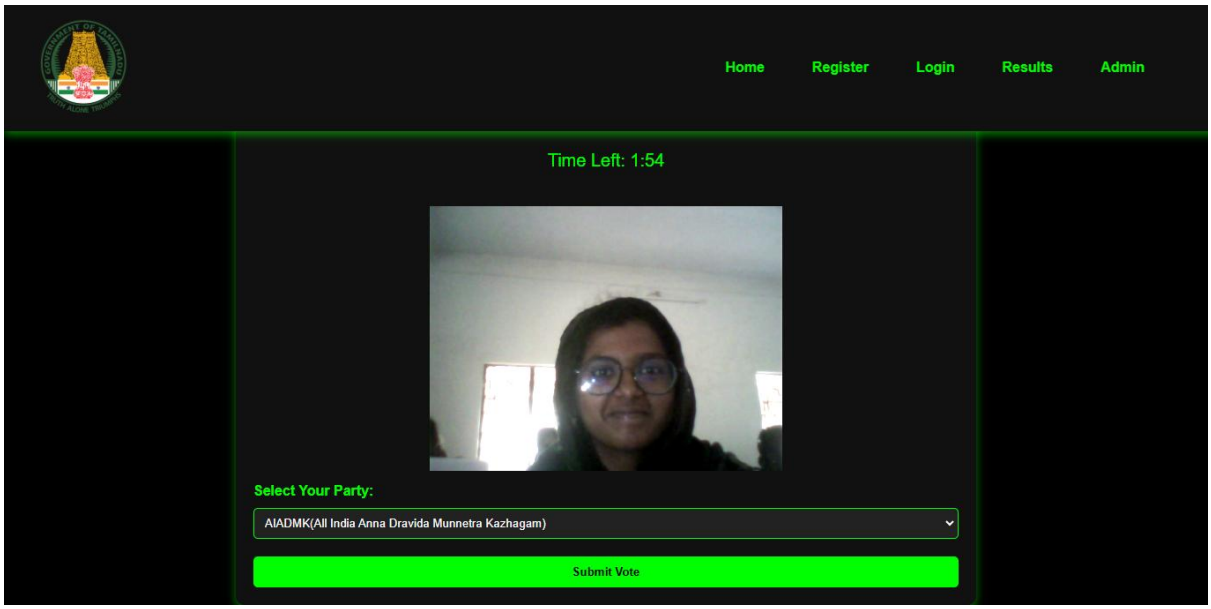
REGISTER PAGE:

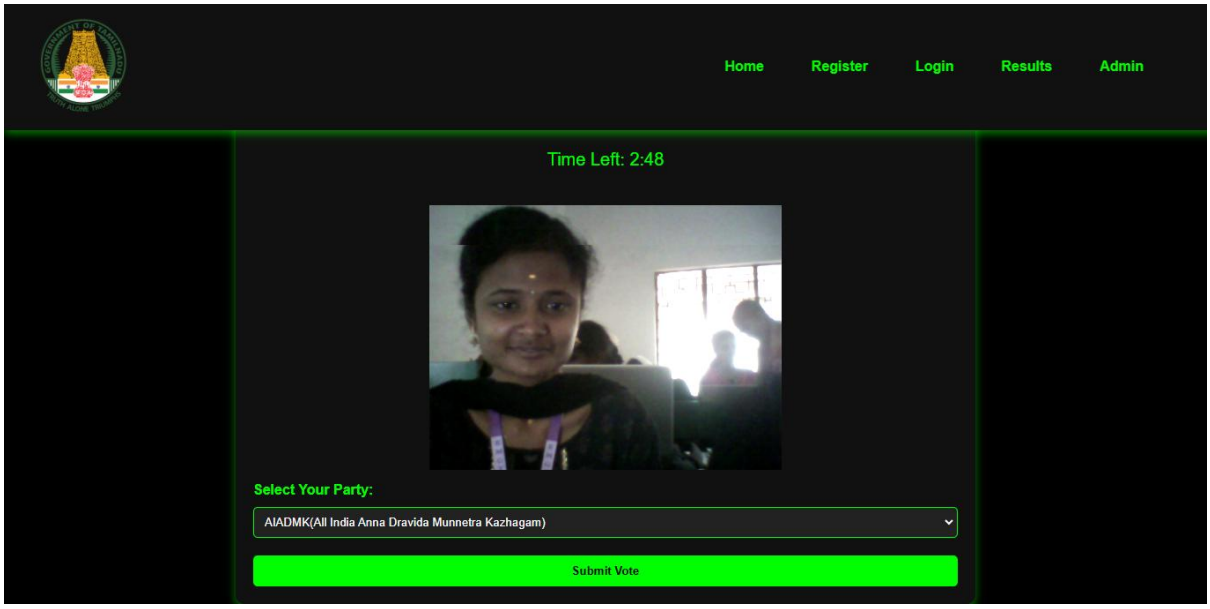


LOGIN PAGE:

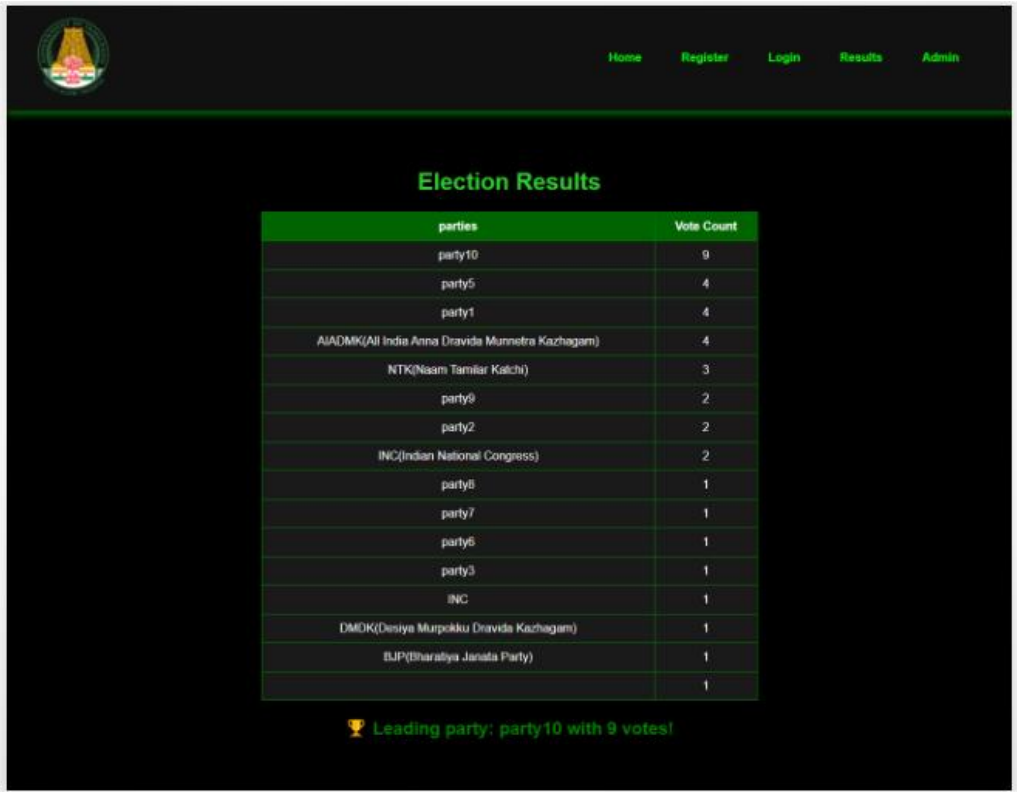


VOTING_PAGE:


Home
Register
Login
Results
Admin
Time Left: 1:54
Select Your Party:
AIADMK(All India Anna Dravida Munnetra Kazhagam)
Submit Vote


Home
Register
Login
Results
Admin
Time Left: 2:48
Select Your Party:
AIADMK(All India Anna Dravida Munnetra Kazhagam)
Submit Vote

RESULT PAGE:


Home
Register
Login
Results
Admin
Election Results
parties
Vote Count
party10
9
party5
4
party1
4
AIADMK(All India Anna Dravida Munnetra Kazhagam)
4
NTK(Naam Tamilar Katchi)
3
party9
2
party2
2
INC(Indian National Congress)
2
party8
1
party7
1
party6
1
party3
1
INC
1
DMDK(Desiya Murpokku Dravida Kazhagam)
1
BJP(Bharatiya Janata Party)
1
1
Leading party: party10 with 9 votes!

ADMIN PAGE:



FUTURE ENHANCEMENT

The future of blockchain-based e-voting systems is poised for significant advancements, leveraging cutting-edge technologies to enhance security, scalability, and accessibility. **Decentralization and scalability improvements** will ensure the system can handle large-scale elections with millions of voters while maintaining a tamper-proof and transparent ledger. By integrating **advanced cryptographic techniques** such as zero-knowledge proofs (ZKP) and homomorphic encryption, voter privacy will be further safeguarded, ensuring that votes remain anonymous while still being verifiable.

With the rapid evolution of **Artificial Intelligence (AI) and Machine Learning (ML)**, AI-driven fraud detection mechanisms will be incorporated to analyze voting patterns and detect anomalies in real-time, preventing potential cyber threats and election manipulation. **Biometric authentication** methods, including facial recognition and fingerprint scanning, will add an extra layer of voter identity verification, reducing the risk of fraudulent voting while ensuring accessibility for all users.

Another key advancement will be the **integration of self-sovereign identity (SSI) systems**, allowing voters to authenticate themselves securely without relying on centralized databases, enhancing both security and privacy. Additionally, the **implementation of mobile and offline voting solutions** will ensure inclusivity, enabling voters in remote areas or with limited internet access to securely cast their votes via blockchain-enabled mobile applications.

Cross-border voting solutions will also be a crucial enhancement, empowering expatriates and overseas citizens to participate in their home country's elections without relying on embassies or postal voting. This will be made possible through blockchain-based identity verification and **multi-signature authentication** to prevent unauthorized access.

To further improve usability, **multi-modal interfaces** will be introduced, allowing voters to interact with the system using voice commands, screen readers, and adaptive technologies, ensuring accessibility for people with disabilities. Moreover, enhanced **real-time auditability** will provide election observers, candidates, and citizens with transparent insights into the voting process, fostering greater trust in electoral systems.

As blockchain technology matures, **post-quantum cryptography** will be integrated to future-proof the system against emerging quantum computing threats, ensuring the long-term security of encrypted votes. Governments and organizations will also adopt **hybrid blockchain architectures**, combining public and private blockchains to balance transparency and privacy, ensuring secure yet verifiable elections.

These advancements will collectively transform blockchain-based e-voting into a **highly secure, transparent, and universally accessible** electoral solution, reshaping democratic processes and ensuring **trustworthy, tamper-proof elections** for the digital age.

CONCLUSION

The **"Blockchain-Based E-Voting System"** revolutionizes the electoral process by leveraging the power of blockchain technology to ensure secure, transparent, and tamper-proof voting. This system enhances the voting experience by eliminating traditional challenges such as electoral fraud, vote manipulation, and unauthorized access. Through decentralized ledger technology, every vote is securely recorded, immutable, and verifiable, ensuring complete trust in the electoral process.

By integrating **smart contracts, cryptographic security, and real-time vote tracking**, this system guarantees fair and accurate election results without the need for intermediaries. Voters can participate in elections from anywhere while maintaining their anonymity, making the system accessible, user-friendly, and efficient. The use of blockchain ensures that once a vote is cast, it cannot be altered or deleted, reinforcing the integrity of the election.

With continuous advancements in **blockchain scalability, cryptographic security, and biometric authentication**, this e-voting system has the potential for further enhancements. Future integrations may include **AI-based fraud detection, multi-device compatibility, and enhanced voter verification mechanisms** to further strengthen the security and accessibility of digital elections.

Ultimately, the **Blockchain-Based E-Voting System** redefines the future of digital democracy by providing a **secure, transparent, and efficient** method of conducting elections, ensuring voter confidence and upholding the fundamental principles of democracy in the digital age.

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